MYP4 Unit 1 Design Summative Portfolio

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SECTION A: INQUIRING & ANALYSING:

Problem Statement
Indian citizens have limited information regarding vaccination, negatively affecting the health of the population.

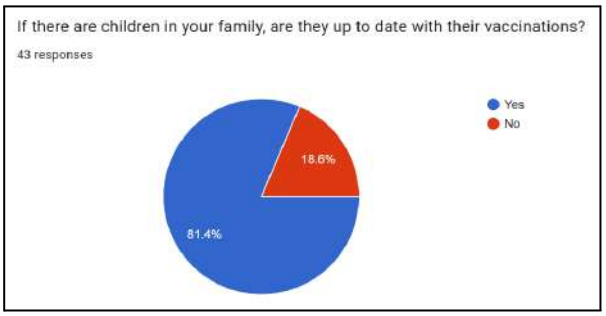
Target Audience
Indian adults and children's caregivers

The Problem
A large part of the research comes from this survey: Vaccination Awareness

What is the problem?

The problem is a lack of clear, organised and understandable knowledge about vaccinations available on the internet.

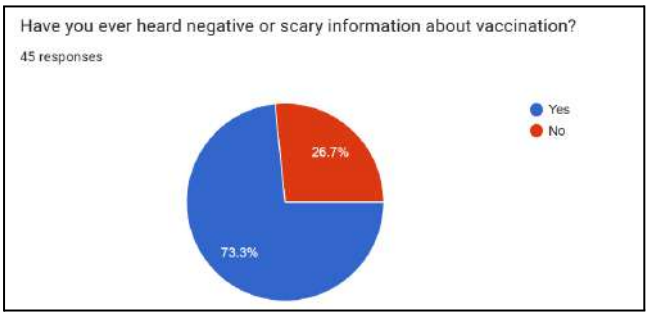
India's basic immunization rates (Basic vaccinations given to younger children) are about 93% with 7% being not fully vaccinated or unvaccinated (PIB). This 7% includes over a million children that are unvaccinated or partially vaccinated. In India, over 500,000 deaths are caused annually by preventable diseases (diseases that could be prevented by vaccination). Results from my survey shows that this 7% reflects not only a lot of the population that comes from rural or remote areas, but also a chunk of the well-educated and urban population, as it shows that out of 43 respondents, all of which come from urban areas from where doctors and vaccinations are very much accessible, over 18% admitted having children in their household who are not up to date with their vaccinations.

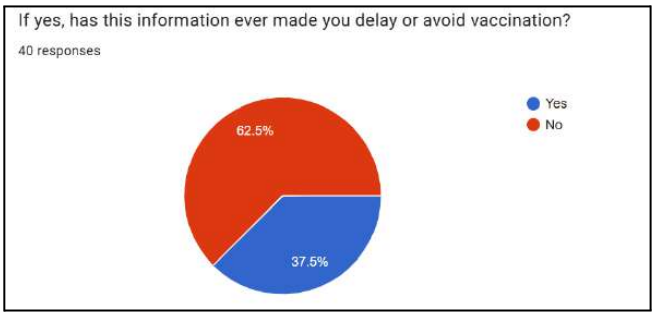


Connecting this to the initial problem statement, according to the WHO, 25% of Indians using social media encounter online misinformation that seeks to discourage them from vaccination. Furthermore, my survey showed that among the respondents 73% have encountered scary information about vaccination online and from word of mouth, and out of them 37% have been discouraged from vaccination in the past due to it. Though there is not much data regarding

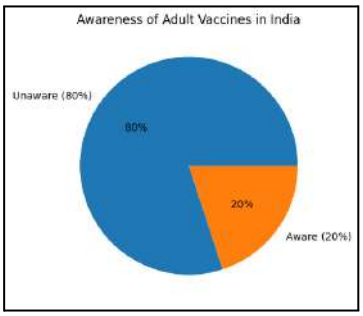
vaccine hesitancy in the status quo, there was a great amount of vaccine hesitancy in India during Covid, with large amounts of people not getting themselves vaccinated despite widespread government vaccination camps where vaccinations were being given for free. (Aggarval, Sumit et al.) I myself knew several people who never took the Covid vaccine because they heard from social media or friends about various concerning side effects.

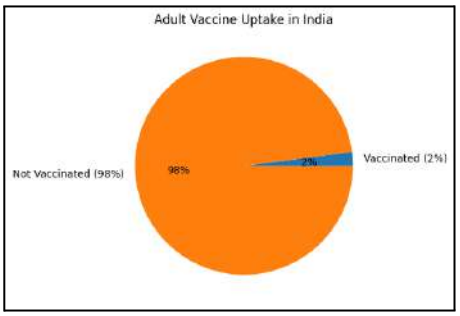
Though there is no equivalent data for India, a UK study suggested that the percentage of people who considered vaccination safe dropped from 90% to 70% post-pandemic due, which is connected to a rise in vaccine misinformation caused by controversies regarding the side effects of the covid-19 vaccine. (Reed) Since reports suggest that even India experienced a surge in vaccine misinformation and anti-vaccine sentiments post-pandemic, (Kumar, Sridevi et al.) a similar drop may as well have occurred in India too.
The WHO even mentioned vaccine hesitancy among the top 10 threats to global public health. (Aggarwal, Sumit et al.) This shows the severity of the amount of misinformation spreading about vaccination, and also that the misinformation was actually successful in stopping people from getting vaccinated.





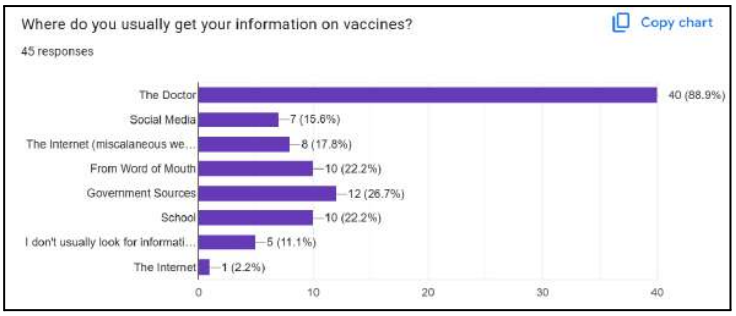
The matter visibly gets even more grave when considering adult vaccinations. Over 80% of Indians are unaware of there being mandatory vaccines meant for adults, with only 2% of that population having taken these vaccines. These frightening percentages are linked to the widespread myth that vaccinations are only for children. (Kulkarni, Taur et al.)



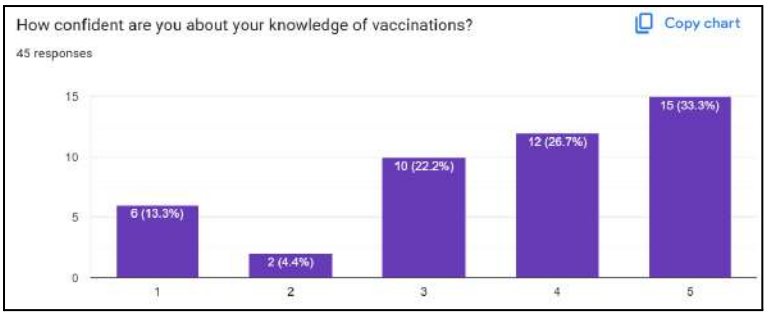


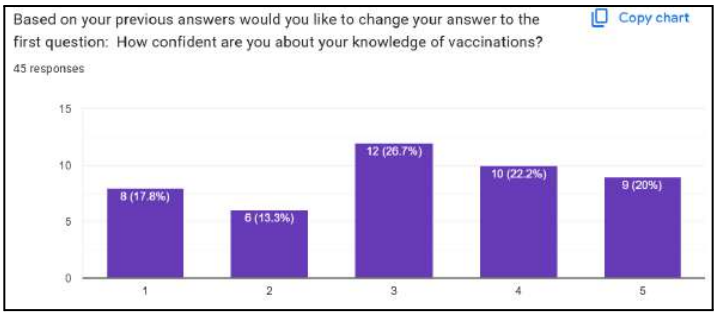
(The above charts are generated by ChatGPT.)

When asked what their primary source of vaccine information was, about 15% replied ‘social media’ and 20% replied ‘the internet.’ The internet or social media are non-credible sources, as literally anyone can go onto the internet and freely upload anything. (Micich, Cross) Additionally, 26% of respondents used government sites as a source of information. The problem with this is that Indian government websites often have severe flaws. For one, several Indian government websites, including vaccine awareness websites, use the outdated ‘http’ rather than ‘https’ which makes the users vulnerable to security risks. (Kanth) I myself experienced a challenge due to this first-hand, when while trying to open an Indian government vaccine awareness website, my browser (Google Chrome) advised me against doing so due to the lack of ‘https.’ If you open the national health mission website, you will get the message that the TSL certificate(a document that confirm that the website you are opening is really what it claims to be and not false) has expired (19 days ago as of 07/02/2026) and that it is a security risk to open the site. This presents security risks for people trying to open these sites. But even if they open these sites, they will face great difficulty finding what they need, as Indian government sites often have cluttered layouts and are extremely difficult to navigate. (Kanth)



What makes this even more dangerous is that despite these terrifying numbers, people are confident about their knowledge on vaccines. In my survey, one of the first questions was ‘How confident are you in your knowledge of vaccines on a scale of 5?’ The most clicked answer was 5/5. However, after a few questions which tested the respondents’ awareness, I gave them the same question again. The second time, the most clicked answer was 3/5. 29% of respondents changed their response. This shows not only the extent of the lack of information, but also that people often still think that they have sufficient information about vaccination.This increases the urgency of the matter significantly.





Now to address a very important point, where does the doctor come into all of this? It is important to note that the doctor should always remain the primary source of medical knowledge. However, in the modern day, people expect to find everything in the digital world. Ever since the 2000s, when the internet became accessible to people, they have started relying more and more on the internet for medical information over the doctor. (Shekhar, Javalkar) Studies suggest that anywhere between 63% to 80% of people use the internet as a source of medical information. (Cohen, Coppola et al.) While this is another entirely different problem to be solved, there needs to be a way to make sure that when people go online in search of medical information, they come across sites that give them credible and correct and complete information rather than spreading misinformation. Repeating the previously mentioned number, 25% of Indian internet users receive vaccine misinformation online.Therefore, until the problem of online dependence for medical knowledge can be stopped, there needs to be a greater number of credible medical websites that present information in an organised, understandable and accessible way while acknowledging and busting misinformation so that people are less likely to stumble upon incorrect information. This mandates the need for the solution to be a digital solution.

All of these examples mentioned above are different aspects of the problem of a lack of credible, organised and understandable information about vaccinations available. The lack of accessible information not only causes serious problems such as the severely low adult vaccination rates but also causes the rising social media misinformation to be believed as it becomes a major source of people's vaccine knowledge, as no online presence of correct information actively tries to combat the online presence of misinformation. This justified a need for a digital platform that gives clear, correct information about vaccines and also acknowledges and busts common myths.

Who is it a problem for?

This is a problem for everyone. Vaccines are important because they prevent deadly diseases such as Covid, Tetanus and Hepatitis. Several of these preventable diseases can spread via surface contact and respiratory droplets, and so are very easily spread and can cause death. (Arora) When people avoid vaccination due to misinformation, it makes

them vulnerable to the disease the vaccine prevents. Considering this, we can see that the problem of vaccine misinformation and the lack of awareness of the importance of vaccines can possibly become a matter of life and death for many people. Hence it can be concluded that this problem impacts the universal health of the public. Furthermore, it also affects them financially. Getting infected by some preventable diseases can result in hospitalisation and long-term health risks. (Unicef) Hence patients of these diseases will need to constantly spend on medical treatment and healthcare. To make this worse, high prices of medical treatment for several preventable diseases in India drives millions into poverty annually. (Phadke) In summary, if someone does not get vaccinated and gets infected by some diseases, they can either get treatment and be driven into poverty or avoid treatment and ruin their health. Even if they choose the second option, they will be at risk of death and may not be able to work and generate income. Hence, either way, this problem affects the public's collective health and economy.

Where is the problem occurring?

The problem is occurring all across India: From villages to cities. However, while government campaigns focus on rural areas, there are also people who live in urban areas who get misinformation about vaccination online, with a 2024 WHO report finding that 25% of Indians have encountered misinformation regarding vaccination online, resulting in anti-vaccine sentiments or misinformation regarding details, effects and necessity. Additionally, many Indians receive circulating misinformation from friends, which may affect their mindset towards vaccination. (Stuedzle, Willis et al.) Even the survey I conducted showed family and friends and social media as the largest contributors to the circulation of discouraging information about vaccination. To make the situation worse, the global public mindset towards vaccinations has worsened following major controversies about covid-19 vaccines. (Arora, Singh et al.) As mentioned earlier, this misinformation discourages people from vaccination, causing them to lose health, as well as money for treatment. The problem is not only occupying space-wise the entirety of India, but the misinformation spreading like wildfire through social media and running deep into the mindset of people.

What is the cause of the problem?

The cause of the problem is a lack of awareness, along with misinformation that spreads through word and online. Existing solutions are government and medical websites that use official or medical terminology that is extremely hard for a common user to interact with. Government websites use large blocks of text which make it difficult for users to navigate for particular information and due to often having expired TLS certificates, they often show as 'http' websites rather than 'https' websites and are shown by browsers as being potentially fraudulent. (Kanth et al.) This rules government websites out. The other solution for people is the internet and social media, which often has heavy misinformation as anyone can add information on them. (Wynn, Skaffle et al.)

What effect is the problem creating?

The problem is reducing the health of the Indian population. Vaccines are important because they immunize a person against deadly diseases such as polio, measles, hepatitis A and B and covid-19. Missing vaccinations severely increases the risk of these diseases, which can lead to hospitalisation and long-term health risks. (Unicef) This not only severely impacts the health of the people, but also their financial wellbeing as treatments for several preventable diseases can be extremely expensive. (Phadke) Hence not being vaccinated can be a scenario of life and death. This, as mentioned earlier, not only affects the health of people, but also the health of the economy and of society, which adds to the urgency of this all-pervading problem.

Action Plan

Research Questions

These research questions will guide my overall research.

- ➔ How severe is the lack of clear information about vaccines in India?
- ➔ Is a solution to the problem needed?
- ➔ What are the gaps in the current solutions and what can be improved about them?

➔ What should the layout of the solution be like?

Research Steps

The below steps are all broken down from the above research questions

1) Justifying the need for a solution. (presented in the section “The Problem’)
    a) Getting a broad picture of the gaps of knowledge regarding vaccination among the Indian population via secondary research.
    b) Create and send out a survey that investigates with its target audience regarding awareness of basic vaccinations, vaccinations for adults, other basic information and any problems caused by the lack of accessible information about vaccination.
2) Identifying the necessary features of the solution.
    a) Interpreting from that survey the precise problems and needs of the target audience.
    b) Speaking with a professional website designer to get inputs on how my website should be designed visually and layout-wise.
    c) Setting precise requirements/key features that existing or future solutions to the problem should have.
3) Analysing similar existing solutions.
    a) Identifying at least 4 existing platforms that have similar functions.
    b) Identifying the gaps in the functions of the platforms by checking if they have the set key features.
4) Composing a design brief summarising the above steps.

From these research steps I aim to gather enough information about the problem and its solution to create specifications for my solution.

Sources

Primary Sources:
- ➢ People who filled the survey
- ➢ Professional website designer

Secondary Sources:
- ➢ Articles from government websites that specialise in medicine
- ➢ Articles from government websites that specialise in statistics
- ➢ Articles from non-government websites that specialise in medicine
- ➢ Research reports from credible organisations or universities

Detailed Research and Analysis Plan

Step	Type	Purpose	Priority
Research			
Research on the severity of the problem of the lack of awareness about vaccination. (presented in the section ‘The Problem’)	Secondary, Qualitative, Quantitative Research	To create a problem statement and justify why the problem is relevant and needs to be solved.	1st priority, since it forms the base of my research and proves the project’s relevance.
Research on why awareness of vaccination is important. (presented in the section ‘The Problem’)	Secondary, Qualitative Research	To create a problem statement and justify why the problem is relevant and needs to be solved.	1st priority, since it forms the base of my research and proves the project’s relevance.
Creating and sending out a	Primary,	To find further data about the	2nd priority, since it will help me

survey that finds data on the problem and the need for a solution.	Qualitative, Quantitative Research	problem and to identify necessary features that a solution must have, which will help me create design specifications.	gain further knowledge about the problem and will form the basis for the 3rd priority step.
Speaking with a professional website designer to gain insight about the visual design and layout of my solution.	Primary, Qualitative Research	To gain professional insight to define some of the visual style and attributes of my solution, which can be turned into design specifications.	4th priority, since the layout and visual style of the solution can only be discussed once the required features of the solution, which the other steps are working towards, have been finalised.
Identifying 4 of the best existing solutions to the problem.	Secondary, Qualitative Research	To find good examples of solutions to the problem that I can learn from by analysing them.	3rd priority, since analysis of these solutions will be dependent on findings of the 2nd priority step.
Analysis			
Analyse the features of 4 similar existing solutions based on the necessary features identified earlier.	Secondary, Qualitative Research, Analysis	To look at existing solutions to take inspiration from them and identify the gaps within them, which will help me create design specifications that make it so that my solution covers those gaps.	3rd priority, since analysis of these solutions will be dependent on findings of the 2nd priority step.
Summary			
Finally, based on all this research and analysis about problems and solutions, I will create a design brief.			

Professional Insights

Since I do not hold special expertise in the designing of a digital platform, to determine some of the attributes the layout of a digital vaccine awareness solution should have, I had a discussion with a professional web designer, Afshan. I first told him the problem I am working to solve, as well as the solution I am aiming for. Below are some key points of the discussion.

The reason a large part of the discussion was centered around government sources was because based on the survey I conducted, among digital solutions, government sources were the most commonly-used by respondents, with 26% of respondents choosing government sources as their source for vaccine awareness. Seeing the popularity of government sources coupled with the severe flaws they have, which were discussed in the section ‘The Problem’, made me have government sources be a large part of the discussion.

👉 How can government websites be improved?
They can improve by splitting large blocks of content into smaller sections, so that people can easily find whatever they are looking for. They have very few visuals and too much text. They should start using more visuals or at least have large titles and good spacing to help the user differentiate different parts of the information pages. Even a contents section on each page helps a user navigate through it. Perhaps, they can have short versions of texts and have a read more button. Additionally, these websites often have expired TLS certificates, which means that they could cause security threats for their users.

👉 What impact does it make on the user?
If users aren’t able to find information on government sites, which are supposed to be credible, they will go to other websites or even social media, which may not be credible. Of course, there is also the angle of security risks due to

expired TLS certificates. These documents make sure that the website is actually what it claims to be and not an imposter that might steal your data. Hence it is dangerous to open websites with an expired TLS certificate as they could be fraudulent sites.

👉 What kind of digital platform should I make, app or website?
For awareness, a website would be better, since the website will be found by people by a simple google search, while an app has to be found by word of mouth or advertisement as people don't scroll on app stores. Hence, to make the platform easier to find you should make it a website.

👉 What fonts or colours should I use?
Have a neutral font that would not feel overwhelming to anyone, like the basic aerial font. For colours, use calm colours so that the site feels 'safe' such as light blue and white. Avoid using red or overly bright colours as they signify danger or activity, which would not fit your health platform.

👉 Is there anything else that could improve the overall solution?
Perhaps FAQ sections or feedback sections. Every platform should have a feedback section as it tells the people running the website about any problems the users might be facing. At least when making a website on an amateur level you might miss things. So a feedback section is necessary.

☑️My major takeaways from the discussion were:

- ➢ Government websites often feel overwhelming due to a poor layout that has extensive blocks of text and few visuals or subheadings. That kind of layout is something I should avoid.
- ➢ The platform should have a basic font like aerial and a calm colour scheme that invokes safety and would not seem overwhelming.
- ➢ The platform should avoid overwhelming fonts or colours.
- ➢ The platform should have an FAQ (frequently asked questions) section (which I had already planned) and a feedback section, since these are necessities for digital platforms to have, especially those that give information.

Key/Necessary Features for the Solution

Based on the previously done primary and secondary research, I will now highlight what could be some key features the final solution should have and why. These key features will be expanded further to make design specifications for the making of the solution.

Key Feature	Purpose
Covers vaccines for all age groups	Since only 2% of India's population is estimated to know about the existence of certain vaccines that are recommended for all adults to take, it is necessary to put some emphasis on vaccines for adults alongside vaccines for children.
Organises vaccines based on mandatory, conditional and optional	This feature is necessary since one of the most common myths about vaccines is that some vaccines that the government claims to be mandatory are actually not necessary since some diseases are not as prevalent in the country, not understanding that disease outbreaks can even be caused if it is carried to India from another country. (Reliance General Insurance) Therefore it is important for a vaccine awareness solution to clarify to the user what vaccines are mandatory, what not and why.
Mentions booster doses	This feature is necessary in order to make a solution a go-to vaccination information solution for users. Though booster doses are not something my research tells me people often struggle with, they should be included in vaccine awareness solutions for the convenience of the user. Research suggests that people often prefer to use an 'all-in-one' digital platform which caters to maximum of their needs in a particular domain over using multiple specialized platforms in order to minimize the cognitive load caused by juggling

	through multiple platforms. (Tatol)
Mentions common allergies (if any)	These features are necessary since one of the most common myths about vaccines is that they have extreme side-effects such as causing autism or infertility. (Dr. Singh) Therefore, it is extremely important for a vaccination awareness solution acknowledge the actual side-effects (which are much more minor for most vaccines such as mild swelling and redness and vaccinated area) as to make it clear to the user what the actual side-effects/allergic reactions are and that they are not nearly as extremely as common myths suggest.
Mentions common side-effects (if any)	
Shows vaccine schedules	This feature is necessary in order to make a solution a go-to vaccination information solution for users. Though vaccine schedules are not something my research tells me people often struggle with, they should be included in vaccine awareness solutions for the convenience of the user. Research suggests that people often prefer to use an 'all-in-one' digital platform which caters to maximum of their needs in a particular domain over using multiple specialized platforms in order to minimize the cognitive load caused by juggling through multiple platforms. (Tatol)
Answers common questions	This feature is necessary in order to answer any of the common questions people have about vaccination. As shown by my survey, one of the common difficulties faced by people while using current solutions is contradictory information, which causes them to question what information is correct. Hence, it is necessary for such a solution to acknowledge common questions like these and answer them.
Busts common myths	This feature is necessary because people are often exposed to a large amount of vaccine misinformation. The WHO identifies misinformation as one of the top ten threats to public health. (Aggarwal, Sumit et al.) There are also many vaccination myths in India that are extremely common, such
Has search functionality	This feature is necessary so that a user can easily find the page for a particular vaccine. I realised this to be an important feature while comparing the several existing solutions in terms of how they present individual vaccines in order to find these key features. For one test, I chose one particular vaccine, the tetanus vaccine, and compared the tetanus vaccine pages on several existing vaccine awareness solutions. But in all of them, since there was no search bar that would let me directly enter the name of the vaccine I wanted to find information about, I had to go through long lists of vaccines to find the tetanus vaccine from them. Therefore, for users who need to find information on a particular vaccine, it is necessary to include a search bar that allows them to go to the page of any vaccine without scrolling through long lists.
Cites sources	This feature is necessary in order to add credibility to the site. Since, as established by my previous research, misinformation about vaccines is extremely common, some users might want to confirm the credibility of a vaccine awareness platform by checking the credibility of the sources of its information.
Has a clean, readable layout	This feature is necessary as it is necessary for any digital awareness platform to not have any clashing text in its layout in order to make it readable.
Allows users to add feedback for the solution	This feature is necessary because users should be able to tell the creators of the solution about any possible problem they might be facing with it so that it can be fixed.

Similar Pre-Existing Solutions

Among the existing solutions I selected for analysis, I wanted to have a spread of different types of solutions, such as official government or WHO websites and non-government initiatives. The reason for analysing different types of sites was so that if these different kinds of sites were doing some things differently between each other, I would have more options to take inspiration from.
Therefore, I chose 2 government websites, nhs and cdg, 1 WHO-associated website, Vaccine Information, and 2 non-government initiatives, Tata 1mg and Immunize India.

Solution Name	Solution Description
Tata 1mg.com/Vaccinations	Tata 1mg is one of India's leading digital healthcare platforms which contains information regarding all kinds of topics relating to medicine, including vaccinations. Note: Even though the vaccination section of this site is labelled as in-progress, the main page and format of said section is publicly visible. It is from that that I will analyse this site.
nhs.UK.Vaccinations	NHS.UK.Vaccinations is a UK government health platform which seeks to spread awareness about vaccinations for children and adults, along with their importance and how to take them.
cdc.gov/Vaccines	cdg.gov/Vaccines is a US government health platform which seeks to spread awareness about vaccinations for children and adults, along with their importance and how to take them.
Immunize India.org	Immunize India.org is an Indian non-profit initiative that seeks to spread awareness about vaccination and improve the nation's immunization rates and health by helping people in rural areas to get vaccinated.
Vaccine information.org	VaccineInformation.org is the awareness platform of Immunization.org, an organisation that promotes vaccination and is associated with the WHO.

Comparison of Solutions

Based on the important features for a vaccination awareness solution previously finalised, below are analysed 4 different existing solutions in a checklist format. The important and necessary features identified form a vision for an ideal solution. By checking how many of these existing solutions actually have those features, I will get a clear idea of the gap between the current solutions and the ideal solution, in which I will try to integrate all of these necessary features. By giving each solution scores in fraction form based on how many out of the 12 features they have will give me a quantitative understanding of the extent to which the existing solutions are close to being the 'ideal solution.' All the below data has been gathered by observation of the platforms. Any additional explanation required is added below the checkbox.

Checklist of Necessary Features in Existing Solutions

	Tata 1mg	nhs.UK	cdg.gov	Immunize India.org	Vaccine Information
Covers vaccines for all age groups	✅	✅	✅	❌ Focuses on only child vaccination.	✅

Organises vaccines based on mandatory, conditional and optional	❌	❌	❌	❌	❌
Mentions booster doses	❌	✅	❌	✅	✅
Mentions common allergies (if any)	❌	❌	❌	❌	❌
Mentions common side-effects	❌	✅	❌	✅	❌
Shows vaccine schedules	❌	✅	❌	✅	✅
Answers common questions	❌	❌	❌	❌	❌
Busts common myths	❌	❌	❌	❌	❌
Has search functionality	❌	❌	❌	❌	❌
Cites sources	❌	❌	❌	✅	❌
Has a clean and readable layout	✅	✅	✅	❌ Text clashes with images	✅
Allows users to add feedback for the solution	❌	❌	❌	❌	❌

Using the above chart I have identified the gaps in the present solutions. As seen in the chart, most of the selected existing solutions don't have most of the necessary features. Below are each of the solutions' scores in terms of how many of said features they have.

- Tata 1mg: 2/12
- Nhs: 5/12
- Cdg: 2/12
- Immunize India: 4/12
- Vaccine Information: 4/12

Some features, such as mentioning common allergies and busting common myths are not present in any of these. This is extremely concerning considering the amount of misinformation regarding the effects of vaccines among the people. Considering the degree of misinformation present, addressing and debunking misinformation and clarifying the actual positive and negative health effects should be a priority. However, very few actually try it.

Design Brief

The following section summarises the initial research and analysis.

Problem Statement

Indian citizens have limited information regarding vaccination policies and regulations, negatively affecting the health of the population.

Target Audience

Indian adults and children's caregivers

Problem Details

- India's basic immunization rates (for basic vaccines given to younger children) are around **93%**.
- Due to India's large population, the remaining **7%** make up over **1 million** children.
- About **500,000** annual deaths occur in India due to preventable diseases (diseases that could be cured by vaccination).
- It is staggering that, as shown by my survey, which was filled by 42 people to date, even among educated and financially well-off households, **17%** of households put the importance of vaccination below 3/5. **20%** of respondents say that their children are not up to date with their vaccinations.
- In the same survey, it was found that **26%** of the respondents claim to have found negative and discouraging information about vaccinations from various sources, mostly social media and word of mouth, and 40% of the people have been discouraged from vaccination due to this.
- Even reports from the WHO suggest that **25%** of people in India find concerning misinformation about vaccinations online.
- The survey also suggested that people face problems with existing solutions to the problem, such as the internet, social media, government sites and sometimes even doctors, such as a lack of updated information, contradictory information and a lack of clarity on side effects.
- **63-80%** of people use the internet as a source of medical information and people are increasingly becoming more and more dependent on the internet for medical information. Hence the solution needs to be digital.
- In summary, people have a difficulty finding the information they need on vaccines and this often discourages them against vaccination or reduces the importance of vaccination in their understanding, putting their and their family's health at stake. The relevance of the problem is that it attempts to improve the health of the country. This is important since if the people are healthy, not only will they live better lives, but the solution will bring us a step closer to eliminating the current problems of society, allowing us to focus on eliminating more of them to improve public health and societal wellbeing.

Solution Requirements (Pre-Specifications)

From primary and secondary research, I have understood the problem of the target audience and created the following requirements. These will be expanded into design specifications that paint a detailed picture of the solution.

- Covers every kind of vaccine, such as mandatory ones for adults and children, as well as conditional or optional ones, along with the reasons for them being mandatory or optional.
- Gives the user freedom to navigate through vaccines in multiple ways, such as by age, by necessity, by effectiveness, etc.
- Gives information about the disease and its severity, as well as the vaccine, such as booster doses, schedules, necessity, consequences of taking or not taking, where and how to get it, common side effects, common misconceptions and why they are false and other things that should be known.
- Information should be arranged in a way that looks approachable and not intimidating.
- Answers common questions.
- Easy-to-navigate layout.
- Allows users to submit feedback about the website.
- Is updated with every major change in vaccines.
- Encourages the user towards embracing immunization.
- Acts as a complete database and the user's one-stop solution for all vaccine information needs.

Current Solutions

Most existing solutions do not have the necessary attributes mentioned above, creating a very large gap between the ideal vaccine awareness platform and the ones that already exist, with none of the existing solutions analysed, out of which were government websites, non-government websites and a WHO-associated website, scoring above 5/12 in terms of how many of the key features it has. It is this gap that I seek to fill by creating a solution that embodies all the key features described above.

Solution Goal

To make a free, easy to navigate website that gives the user all the information on vaccines a person would need and

would answer all doubts and questions, encouraging immunization and would also acknowledge and bust common misconceptions.
Conclusion
In design, we aim to constantly solve problems to make the world a better place. It is solutions like these that make great impacts and improve the quality of life for millions. By making this platform I aim to change people's views on vaccination from potentially dangerous and hard to find information on to a blessing to health and easy to find information on. Throughout this unit, I will consider the needs of society and create a solution that is tailored to them.

SECTION B: DEVELOPING IDEAS:

Design Specifications			
Specification name	**Specification description**	**Justification**	**Success criteria**
Visuals			
Arial font	The font used throughout the website must be arial.	The arial font is pre-installed in all devices and is considered to have high legibility in a variety of different resolutions, making it a safe font to use for any medium. (Matilda) This will make sure the website is readable.	The website must have arial as a universal font.
Colours	The colour scheme of blue and white must be used throughout the website.	The colour blue evokes safety, surety knowledge and professionalism and the colour white evokes peace and clarity. (Rodin) Hence, using the website, the user will feel safe while reading and will feel the website is professional and credible.	A design philosophy expert must confirm that the colours evoke the same emotions as I claim.
Language and Tone			
Medical vocabulary	The website must avoid using medical lingo if common vocabulary can be used instead. For specific situations such as disease names, descriptions, etc, where it would be necessary to use medical vocabulary, concepts will be explained in common vocabulary as well.	Medical and scientific communities suggest that websites conveying medical information to a non-medical audience should employ strategies such as avoiding medical terminology if common terminology is available and following difficult words with explanations. (Coleman, Turacoz) This is necessary for my website, as it is made for the common Indian public, most of which is not of a medical background.	The language must be understandable to my target audience, Indian citizens, and should also be approved by a medical professional as adequate.
Credibility			
Citations	Each page of the website must cite its	Citation makes a website more credible as it makes the sources accessible for someone	All sources used must be cited.

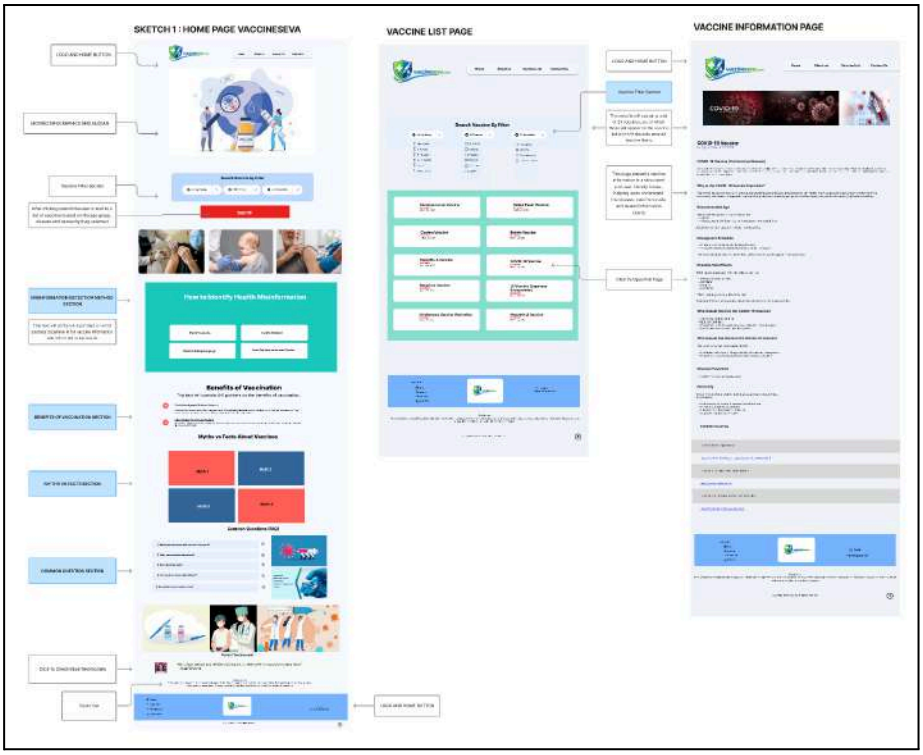
	sources.	wanting to verify the content. (MIT Libraries) Therefore, the use and mention of credible sources is crucial to assure a skeptical user that the data provided in the website is genuine. Furthermore, my research showed that a large reason for the spread of vaccination misinformation is a lack of credible sources, which is more reason to include citations.	
Last update date	Each page of the website must show the date it was last updated on.	Adding the date of the last update helps users judge the credibility of the data mentioned in the website based on its age, as older pages may contain outdated and misleading information. Furthermore, as found by my research outdated information can be a major medium for the propagation of misinformation. Therefore, it is crucial to show the date of the last update on each page to keep the audience's trust. (Tsytrosh)	Each page must have the last update date mentioned.
Disclaimer	The main page and vaccine pages must display a disclaimer that tells the user to verify the information read on the website with a medical professional before taking any step based on it.	The disclaimer ensures the safety of the user, making it clear that the information provided on a medical website cannot act as a substitute for professional medical advice or diagnosis. (Way)	The disclaimer must clearly express the idea that this website cannot replace professional advice or diagnosis.
Navigation			
Search bar	There must be a search bar which takes the user to the vaccine's page when its name or the name of its disease is searched.	A search bar helps the user find a particular page they want to visit without going through lists. So a user who needs to find information on a specific vaccine can use the search bar to reach its page rather than going through organised lists.	The search bar must be able to show the closest results when the name of a vaccine or disease is searched.
Navigation options	The main page must have a filter system which allows the user to navigate through personalised lists of vaccines by entering information such as age group, disease and necessity of the vaccine.	This is necessary so that users with different needs can navigate the website in the way that suits their requirement best. As seen in the survey, most of the audience voted for including multiple navigation options in the website when asked which navigation options they would prefer (age group, disease, necessity, other). Such a filter system will fulfill this.	The different navigation options must work and show a list of vaccines that fit the information entered (filters applied).
Vaccine Pages			

Vaccines included	The website will include all vaccines that are part of India's universal immunization program, travel vaccines, situational vaccines and other categories that my research would tell are important.	The mention of all crucial vaccines will make the list complete. It has been made clear prior that my website is meant to be a one-stop solution for all vaccine information help. Therefore, it is important for it to have a complete list of vaccines.	The list of vaccines is approved by a medical professional as complete.
Information	Each vaccine page must include the disease the vaccine protects against along with its effects, severity and how it spreads.	This will inform the user of how severe the disease is and hence how important taking the vaccine is.	The information must be sourced from credible sources such as government websites or websites with medicine as their primary domain. Said sources must be from the past 3 years and should have the 'https' heading.
Information	Each vaccine page must mention the booster doses and when they must be taken.	Users will stay informed about when booster doses are required, helping them maintain long term immunity.	
Information	Each vaccine page must mention the groups who the vaccine is advisable for and those it is not advisable for.	Users can identify if a vaccine is suitable for them or their family members.	
Information	Each vaccine page must acknowledge and debunk misconceptions about the vaccine (if any).	My research showed me that misinformation is a major factor contributing to people avoiding vaccination. Therefore, the solution must solve that problem.	
Information	Each vaccine page must mention the side-effects or allergies associated with the vaccine (if any).	My research shows that some of the most common vaccine misconceptions are related to allergies or side-effects. Therefore, it is necessary to clear doubts users may have regarding this.	

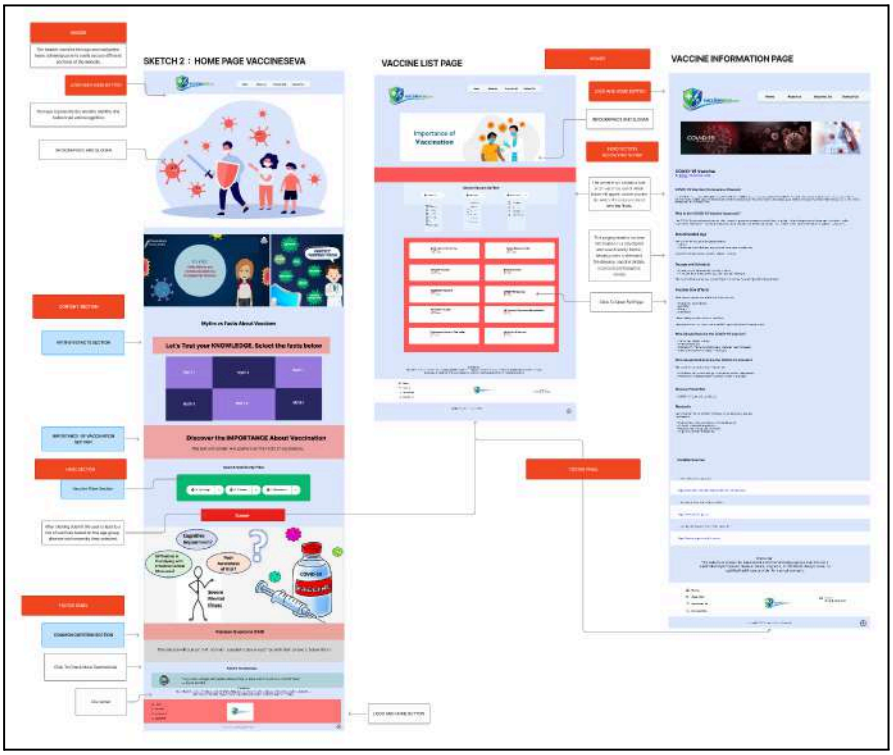
Sketches

Below are 3 different sketches showing variations of the design described in the specifications.
Note: Please open PDF files or Figma files provided if images are not clear.
FigmaFile:https://www.figma.com/board/W8AFk29H5MRjqQ6CqF0ppm/VACCINESEVA-WEBSITE-FIGJAM-DESIGNS-BY-NILAX?node-id=0-1&t=k7ZZoDa0YnZ0regz-1

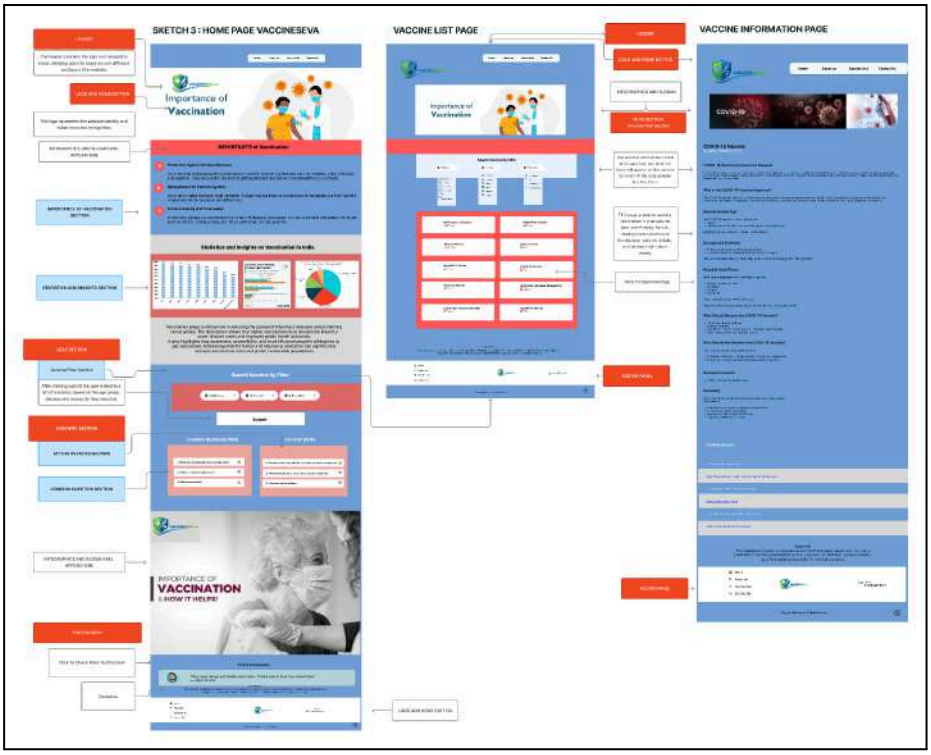
Design 1: 👉
Emphasis on formal feel, navigation and information
PDF: https://drive.google.com/file/d/1S0VH0QOMmNHKi4sTljE9-OpuUkKVxxST/view?usp=drive_link
Emphasis on Navigation



Design 2:👉
Emphasis on interactivity, engagement and misinformation
PDF: https://drive.google.com/file/d/1R5cPNsp0cipXmJysalwENLv__z2wrolE/view?usp=sharing
Emphasis on Interactivity and Misinformation



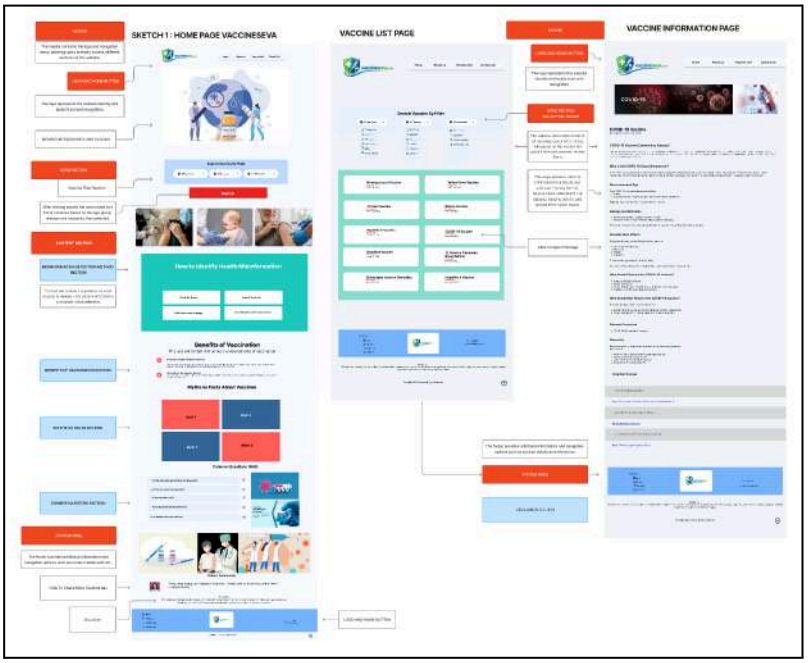
Design 3:👉
Emphasis on formal feel and emphasis on general information of vaccines (importance of vaccination)
PDF: https://drive.google.com/file/d/1ZDdHKbm8FSm6pi-8ElpG2oHSR0bsbJ1J/view?usp=sharing
Emphasis on Statistics



✅Choosing a Final Design

Chosen Design

Design 1✅
PDF: https://drive.google.com/file/d/1S0VH0QOMmNHKi4sTljE9-OpuUkKVxxST/view?usp=sharing

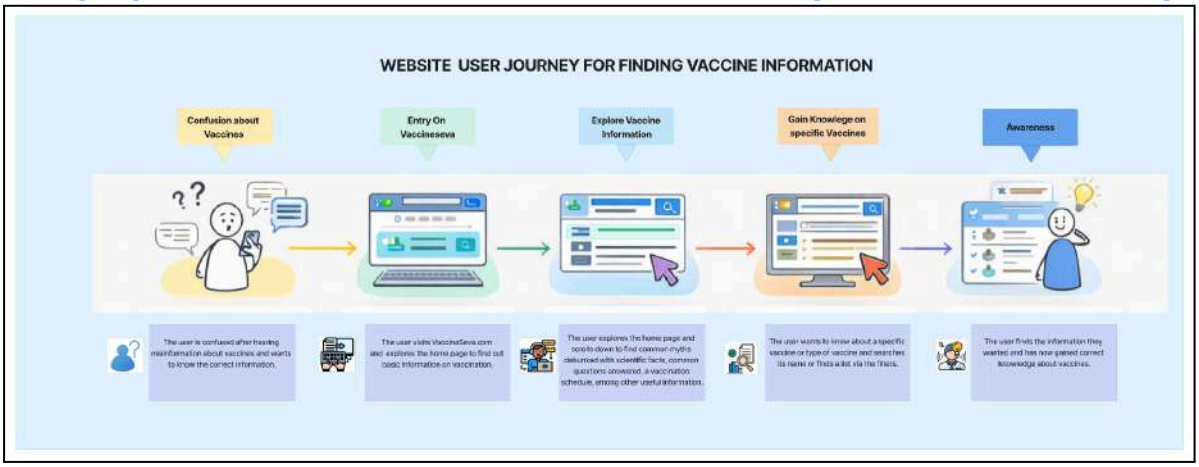


✅ Reason for selection

The three designs feature the same content, the only difference being that different aspects of it are prioritised in each of them. The chosen design keeps navigation elements first, followed by sections on common myths and the importance of vaccines. This means that the vaccine information pages (accessed via the navigation buttons), which serve the primary function of the solution, will be the first thing that the user sees.
Designs 2 and 3 prioritise misinformation on vaccines and the importance of vaccines respectively. This delays the user finding the vaccine information as it is developed further down a page. Furthermore, since the navigation buttons will be small and so will not take up much space, even if they come first as in design 1, they wouldn't take attention away from the other elements. Therefore, design 1 was chosen.

✅ How it will solve the problem: User Journey

PDF: https://drive.google.com/file/d/13-U0xO_d7dMETrSuaaB2bWwJu3g0fmJ8/view?usp=sharing



- A user unable to find credible sources for vaccine information or one who is confused about a vaccine can visit the website to find credible, organised, personalised and easy-to-understand information,
- It presents the navigation options at the topmost, providing the users with the vaccine information they need quickly.
- It acknowledges common myths and questions about vaccines and debunks them. Based on my research, misinformation is a demotivator of vaccinations for many. Hence, a method to acknowledge misinformation in this design is helpful.
- Filtering options such as age group and necessity will allow them to find personalised information.
- The layout is simple and easy to understand, as it clearly splits the page between navigation and general information about vaccines, which will allow the audience to quickly find all the information they would want.
- It fulfills the function of a free, easy to navigate website that gives the user all the information on vaccines a person would need and would answer all doubts and questions, encouraging immunization and would also acknowledge and bust common misconceptions.

✅ How does it fit the target audience

As established prior, the target audience is Indian citizens.
Their requirements that were established prior to making the specifications for the solution were (in summary):
A complete, user-friendly and informative vaccine awareness platform. The solution must include comprehensive information, various navigation options, multiple navigation options and features that promote awareness about both the true information about vaccines and the impact of misinformation. It should be updated and an easy to use source for all kinds of vaccine information.

The design fits the requirements. It meets the specification of being a one-stop vaccine solution by including all information about each type of vaccine, such as its target group, target disease, side effects and myths, along with

general information about vaccines such as their importance and myths. Therefore, the website fits all information about vaccines into itself.

To meet the specification of flexible navigation, the design provides the user with navigation options such as age groups and necessity, which will make sure the user finds personalised information that they need.

The 'myths vs facts' and 'importance of vaccines' sections encourage vaccination and discourage trusting misinformation, making them of use to the target audience, who are often discouraged from vaccination by misinformation. Therefore, they meet the specification of promoting vaccination and discrediting misinformation.

Therefore, the various aspects of the design will help solve the problem of the target audience.

✅ Questions the designer might ask and their answers

The designer might ask:

➢ What font size and style should be used?

Font size used will be medium for text and large for titles.

➢ How will the navigation buttons work (dropdowns, hover)?

The navigation buttons will be dropdowns that display various options. For example, clicking on 'age group' reveals age categories: 6 months, 1 year, 3 year, 10 years, etc.

➢ How does the filter system work?

It works very similarly to the product filters you will find on amazon.com or filters for films on bookmyshow.com. Once the user fills in the filters and clicks submit, they will find, out of a large database of vaccines, the ones that fit their description. For example, if I fill '6 months, tetanus, mandatory', I will get all vaccines for tetanus that are mandatory for 6 month old babies.

➢ What visual elements such as images, icons, real photos or illustrations should be displayed?

Images will include images demonstrating how vaccines protect people

➢ How will the layout change on different devices (phones, laptops)?

The layout will largely be the same, except, some sections will have to be changed in the way they are arranged. For example, some elements that are arranged in a single horizontal line in a laptop (which allows for a larger horizontal view) may be aligned in a vertical line in the phone view (which allows for a larger vertical view).

These responses add further details to the sketch and will be used alongside it to dictate the final design.

✅ What might enhance the effectiveness of the design

Based on the questions that the designer might ask for further clarification, the design could be enhanced by:

- ➢ Specifying and standardising font styles.
- ➢ Showing how the navigation buttons will function.
- ➢ Showing visual elements such as images, posters, icons and the formats for each element.
- ➢ Show versions of each page adapted to different devices and screen formats.

Specification Match

Specification	Match	Evidence
Arial font	✅	The Arial font has been used.
Colours	✅	The colour scheme is primarily white and blue.
Medical vocabulary	✅	I asked a doctor to review the language of the sample vaccine information page I had, to which he approved, saying that it was a good balance of medical and normal vocabulary so that anyone could understand.

Citations	✅	Each vaccine page includes its citations.
Last update date	✅	Each vaccine page shows its last update date.
Disclaimer	✅	The disclaimer is present on the main page and ‘About Us’ page.
Search bar	✅	The search bar is present on the main page.
Navigation options	✅	The navigation filters are present on the main page.
Information - Disease name	✅	Each vaccine page mentions the diseases the vaccine protects against.
Information - Booster doses	✅	Each vaccine page mentions any booster doses required for the vaccine.
Information - Groups vaccines are advisable or unadvisable for	✅	Each vaccine page mentions the groups the vaccine is advisable or unadvisable for.
Information - Side effects and allergies	✅	Each vaccine page mentions the vaccine’s side effects and allergies.

Feasibility

👉 What research is required to build the solution?

The research required to build the solution is:
- ➢ A list of vaccines that would need to be present on the website.
- ➢ Information on each of those vaccines based on the information specifications from credible sources.
- ➢ Common myths, common questions and other information for the main page.
- ➢ Finding a software that will allow me to design and publish the website.

👉 What materials and resources are required to build the solution?

The materials and resources required to build the solution are:
- ➢ A software that will allow me to design and publish the website.
- ➢ A mentor who is an expert at website development with whom I can work to build the solution.
- ➢ Credible medical or government medical sources.

👉 What skills are required to build the solution?

The skills required to build the solution are:
- ➢ Web designing skills.
- ➢ Expertise in whatever software is chosen.

👉 How much time is required to build the solution?

The time available to build the solution after the final design is chosen and process is planned is from the end of march to approximately late april, which is when I estimate the task of building the solution will be due.

👉 How much money is required to build the solution?

The only thing required to build the solution that may require money is buying the domain and hosting for the website. The domain ‘VaccineSeva.com’ costs ₹900 per year, while hosting costs ₹1500. Therefore, according to my current research, the solution will cost me ₹2400 per year if it is to be continued.

👉 What safety concerns might be involved in building the solution?

There are as of now no foreseen safety concerns in building the solution.

👉 Is the idea feasible?

Yes, the idea is feasible.

This is because the required research can be done on easily available websites, such as government websites. Therefore, this will take very less time, 2-3 days at most, and no money. Therefore, this step can be easily completed.

The materials required are minimal. A website development software can be easily accessed on the internet once chosen and a mentor can easily be found by way of contacts. If these two things are found as soon as possible, the rest of the process will go extremely efficiently.

In terms of skill, I would have basic skills in designing a digital solution due to various more basic digital solutions I have designed in the past. The gaps in my skill will be filled by the mentor, with whose help I will also enhance my website development skills while building the solution. In this way, I will be able to easily build the solution in a month. Therefore, in terms of skills, no problem is seen.

Therefore, considering the availability of resources, manageable skill requirements and adequate time, this project is very feasible to execute.

Naming and Logo of the Solution

Before moving onto the final designs, I would like to address the naming and logo of the solution.
I decided to use an extremely simple name so that it would appeal to anyone rather than a specific audience, since my target audience is all Indian citizens. In order to emphasize the fact that the information given would be relevant specifically to India, I chose a hindi name. The name chosen was VaccineSeva.com.
To accompany it, I chose a logo with the motifs of a syringe, for vaccination and a shield, to emphasize how vaccines protect people from diseases.



Final Design

Based on the chosen sketch, I have created a detailed design for all pages, as shown below.

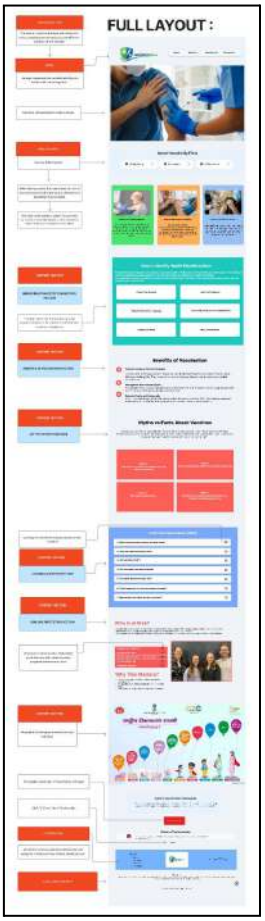
Note: Please open PDF files or Figma files provided if images are not clear.
Figma File:
https://www.figma.com/board/W8AFk29H5MRjqQ6CqF0ppm/VACCINESEVA-WEBSITE-FIGJAM-DESIGNS-BY-NILAX?node-id=0-1&t=k7ZZoDa0YnZ0regz-1

Medium: Figma, FigJam

Full Layout With Annotation

PDF 1: https://drive.google.com/file/d/1zsaNwQjmpUs3yVPq_I258P0x1AYUsIsw/view?usp=sharing

PDF 2: https://drive.google.com/file/d/1wnKcKMfunFtCpVAtcCuPE3XjE_fiLw8j/view?usp=sharing

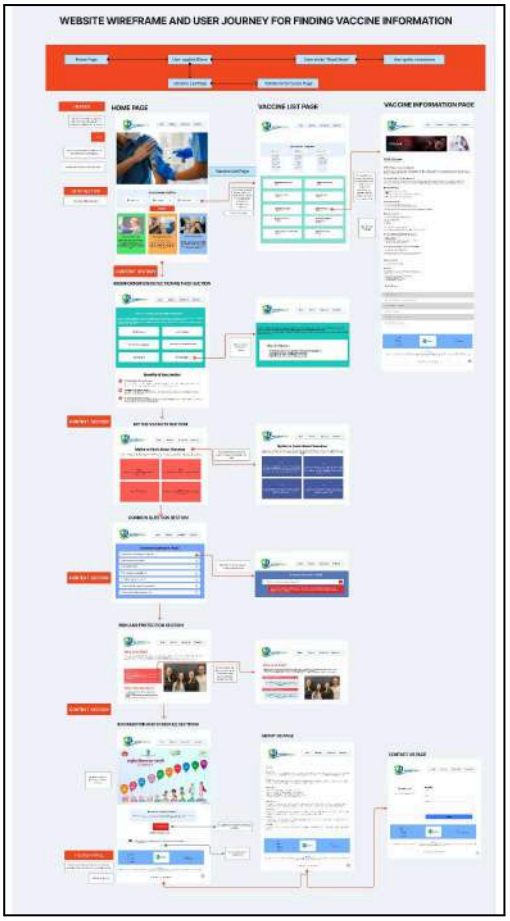




User's Journey and Wireframe

PDF: https://drive.google.com/file/d/1XPbuTzQxGGhgRQRuEH_K0ymPwoTYHfbr/view?usp=sharing
PDF: https://drive.google.com/file/d/13-U0xO_d7dMETrSuaaB2bWwJu3g0fmJ8/view?usp=sharin



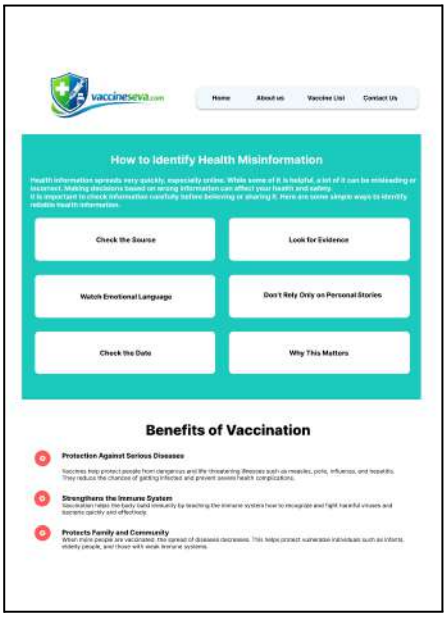


Home Page (Full Layout)

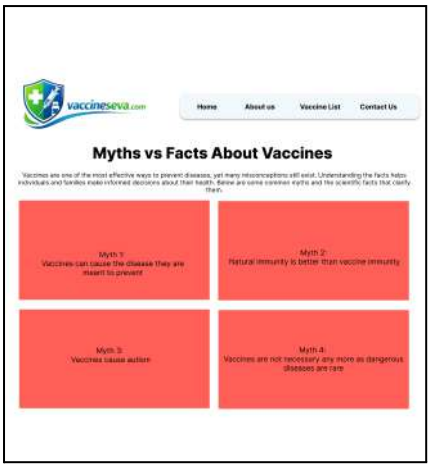
HEADER SECTION , HERO SECTION , CONTENT SECTION AND FOOTER SECTION

PDF: https://drive.google.com/file/d/15GAS82f05ZEgs-QnaUGhHhO3E7lg-VGf/view?usp=sharing













Vaccine List Page

PDF: https://drive.google.com/file/d/1e6TvTOkyRj-7Odk9UJmWlyW9Zwwq78mi/view?usp=sharing



Vaccine Information Page (COVID-19 Vaccine)

PDF: https://drive.google.com/file/d/1_IgiDR4_xqj35JRX7Sf6_mFgqTDABIQV/view?usp=sharing



About Us Page

PDF: https://drive.google.com/file/d/1IxhRD_IZAbtbh3fpah3C-JV8gAWy0wqc/view?usp=drive_link



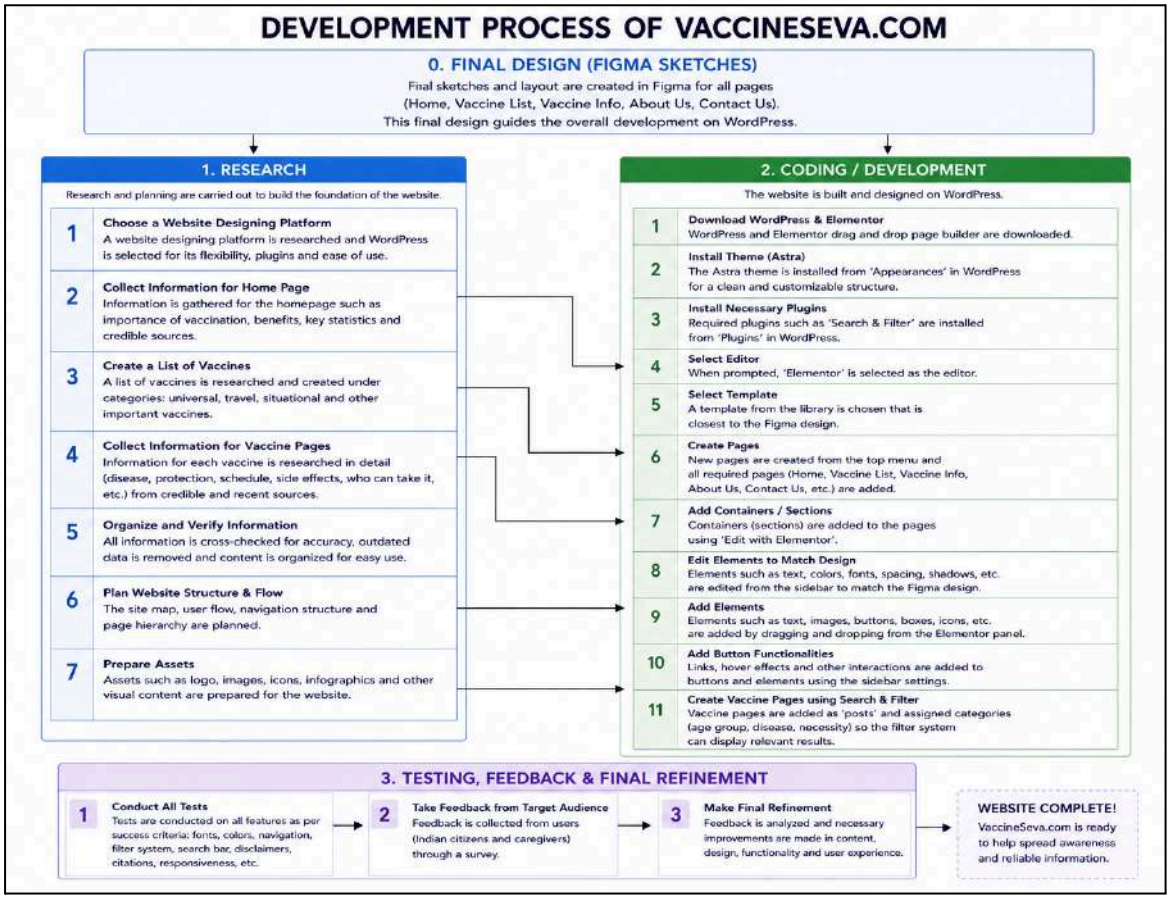
Contact Us Page

PDF: https://drive.google.com/file/d/1tCu7IiBs0bluVgjh1IFD1HLCH0trVEYh/view?usp=drive_link



Project Flowchart

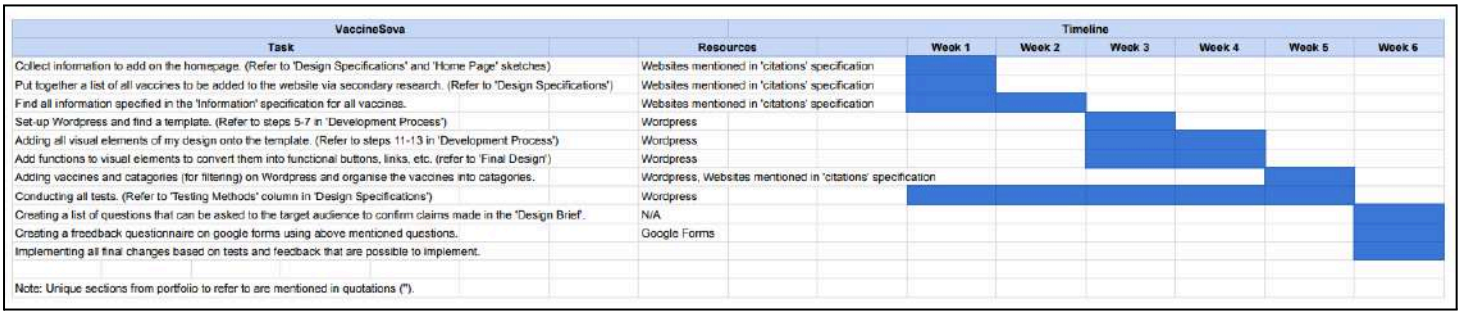
The below flowchart shows the research and development plan after the making of the final design.



SECTION C: CREATING THE SOLUTION:

Gantt Chart

Link: https://docs.google.com/spreadsheets/d/1htNXU3twl70mj2J37gwDDveI5VSO6ELJmaGxrTHmr1I/edit?usp=sharing



VaccineSeva		Timeline					
Task	Resources	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Collect information to add on the homepage. (Refer to 'Design Specifications' and 'Home Page' sketches)	Websites mentioned in 'citations' specification						
Put together a list of all vaccines to be added to the website via secondary research. (Refer to 'Design Specifications')	Websites mentioned in 'citations' specification						
Find all information specified in the 'Information' specification for all vaccines.	Websites mentioned in 'citations' specification						
Set-up Wordpress and find a template. (Refer to steps 5-7 in 'Development Process')	Wordpress						
Adding all visual elements of my design onto the template. (Refer to steps 11-13 in 'Development Process')	Wordpress						
Add functions to visual elements to convert them into functional buttons, links, etc. (refer to 'Final Design')	Wordpress						
Adding vaccines and catagories (for filtering) on Wordpress and organise the vaccines into catagories.	Wordpress, Websites mentioned in 'citations' specification						
Conducting all tests. (Refer to 'Testing Methods' column in 'Design Specifications')	Wordpress						
Creating a list of questions that can be asked to the target audience to confirm claims made in the 'Design Brief'.	N/A						
Creating a freedback questionnaire on google forms using above mentioned questions.	Google Forms						
Implementing all final changes based on tests and feedback that are possible to implement.							
Note: Unique sections from portfolio to refer to are mentioned in quotations ('').							

Development Process

This is the development process after the selection of the final design.
The platform chosen for making the design is WordPress, which was chosen for its unique precoded functionality, which allows developers to add diverse functions to their website with little to no need for coding.

The steps from the below process involving WordPress were written with help from a YouTube tutorial of WordPress by MeticsMedia (view here) and my cousin's friend Zahid, who is a website developer.

Step	Step	Duration	Justification of Time Taken	Success Criteria
		Research		
1	Choosing a website designing platform. (Platform found was WordPress)	1 day	This step will take a thorough discussion with a website development expert and time for thought.	The platform allows me to make the website as it is described in the specifications and shown in the final design.
2	Collecting information to add on the home page.	2 days	This step will contain very little research, much of which will be based on research previously done.	The information comes from Indian government, WHO and recognised medical sources.
3	Creating a list of vaccines to include on the website.	1 day	Since this step is crucial to the website's primary aim, it will include extensive research with credible sources to create a list of vaccines that would be included in the website.	The list is backed by Indian government, WHO and recognised medical sources.
4	Collecting information to add on the vaccine pages.	3 days	Since this step is crucial to the website's primary aim, it will include extensive research and organisation with credible sources to find the most reliable information on each vaccine. Furthermore, adding it on the pages will require a lot of	The information comes from Indian government, WHO and recognised medical sources.

			formatting.	
Setting up WordPress				
5	Download WordPress and the Elementor drag and drop editor.	Momentary	This step is just a matter of downloading two softwares.	WordPress and Elementor are downloaded.
6	In WordPress, go to ‘Appearances’ (side menu) and install a theme (structure for website with functions precoded). The most commonly recommended theme for beginners is Astra, as it provides a wider range of visual and functional possibilities, which I will use.	Momentary	This step is just a matter of installing a large item and will not require any work.	Astra is installed.
7	Go to ‘Plugins’ (side menu) and install necessary plugins for the website. Plugins are tools that help give the website certain pre-coded features. My website would require a Plugin called ‘Search & Filter’, which allows the website to have filter functionality.	1 day	This step will require research and discussion with an expert to find which plugins are required for my website.	Plugins help fulfill all specifications.
Designing on WordPress				
Always click ‘Publish’ on the top menu to save changes.				
8	When the software asks which editor is to be used, select ‘Elementor’.	Momentary	This step is a matter of selecting an option.	The button ‘Edit with Elementor’ is visible on the top menu.
9	From a wide range of templates, select one that is closest to the Figma design.	Within one day	This step will require going through many templates to find the one that would be easiest to adapt into my design.	The template resembles the Figma design.
10	To create a new page, click ‘New’ on the top menu and select ‘Page’. Name the page.	Momentary	This step is the click of a button.	A new page is created.
11	To add elements, click on ‘edit with Elementor’ (top menu). A side menu will appear, from which containers can be added. Containers are sections on a page in which different elements can be added.	Variable	This instruction is a common instruction important to the next 4 steps.	N/A

12	Change elements to replicate the Figma designs: Click on ‘edit with elementor’ on the top menu. Once you click on any element on the website page a sidebar will open with details of the element (text, colour, shadow, etc). Change those details to fit elements of the Figma design.	Total adding and editing of elements will take 3 days.	This step will take a lot of time even if I take the mentoring of a professional, as this would be my first time designing a website on WordPress and would take at least 3 full days for me to complete.	Elements resemble the Figma design and function as mentioned in the Specifications. No elements should clash with one another and should be in format.
13	Add elements to replicate Figma design: Click on ‘edit with elementor’ on the top menu. Then elements such as boxes, text, buttons, etc will appear at the side of the screen and can be dragged and dropped into the website.			
14	Add the functions of different buttons. This is done by clicking on the element, which opens the sidebar. There will be options such as adding a link so that the button leads to a particular page when clicked and settings to set what happens when the element is hovered over.			
15	The vaccine pages would be added via a plugin known as ‘Search & Filter’. This plugin allows the developer to create a set of pages called ‘posts’ which can be organised into different groups or categories. I will add the pages for each vaccine as a post and assign each an age group, necessity and disease.	Within one day	This section only requires copy-pasting research done in the research section. However, it is still a lengthy task as the amount of information to add is very large and would still require a large part of the day.	The filters provide the user with appropriate vaccine lists for any data entered into the filters.
Feedback and Testing				
19	Conducting all the tests.	Done alongside each previous step	Each test will be conducted whenever possible alongside the previous steps.	The final design meets all specifications.
20	Taking feedback from the target audience.	2 days	Feedback will either be collected as an online questionnaire sent to the target audience immediately after	The questionnaire is sent to people from the target audience, the target

			the website is finished, with results collected the next day, taking no more than 2 days.	audience uses the website and is able to comment on all aspects of it.
21	Making final refinement.	2 days	2 days have been kept reserved for any changes that might be done after testing and feedback.	The final design meets all specifications and feedback and changes are justified logically with audience and purpose in mind.

Conclusion

By using the above design and process to make this platform I aim to change people's views on vaccination from potentially dangerous and hard to find information on to a blessing to health and easy to find information on. Throughout this unit, I will consider the needs of society and create a solution that is tailored to them.

Development Process Journal

Below is the process journal of the development of VaccineSeva.com from the initial research required to build it to publishing, and the skills used in it, along with evidence.
The process was largely the same as the process described prior. Below are also screen recordings as evidence of the steps: Process Journal Screen Recordings

Step	Skill Demonstrated proficiency	Procedure and Proficiency
Researching for a list of vaccines	Secondary research skills	I first determined via research what types of vaccines are provided in India, UIP vaccines, vaccines for seniors, teenagers, pregnant women, etc. I researched for and assembled complete lists for each of the categories to make sure that I get a very broad and complete list compared to if I just searched the names of vaccines in general.
Researching for content on the main page	Secondary research skills	To find content such as common myths and common questions I relied on both primary and secondary research. Secondary research on the internet provided me with common data points, while the primary research survey provided me with certain questions that may not be common, but were relevant nonetheless.
Researching for content on the vaccine pages	Secondary research skills	I made sure the websites I used for this were either government websites or websites that have the 'https' header, which signifies legitimate and had medicine as their primary domain. I also made sure that the articles were not older than 2024. I also made sure to track down sources that were research papers with actual primary data, to ensure credibility.
Setting up WordPress	Basic digital literacy skills	This involved simply going on the Wordpress website and clicking a button to download the Wordpress file.
Researching to find appropriate templates	Research and analysis skills	According to common opinion on the internet, as well as my mentor, 'astra' is the best theme to use on Wordpress. Hence, I looked for templates to fit the astra theme. Since I wanted to make sure to execute the next tasks as soon as possible, I looked for templates which had

		features or elements similar to my design on them, such as dropdowns, flipping cards, etc. I chose the one which had the most of the needed features.
Researching to find appropriate plugins	Research and analysis skills	I noted down the various functionalities my website was going to have, such as search, filter, etc. I then spoke to my mentor to identify Wordpress plugins that would help facilitate these functions, which I then downloaded.
Setting up website pages	Basic digital literacy skills	Pages were added with the click of a “+” button and the typing of a title for them.
Adding elements to main page	UI design skills	At the side bar on Wordpress, there are options to add a range of different types of elements. When an existing element is clicked on, the sidebar displays the different attributes of that element instead (text, size, function, etc.) Via this dashboard and also by manually moving elements to different positions, I made the template look and function more like the design for my website.
Adding functions to main page	UI design skills	These functions were added in various ways, mostly by editing attributes via sidebars, such as a dropdown opening when a text is clicked on. Other buttons worked by embedding links in them. For this step I took help from my mentor to learn how different attributes of elements were changed.
Creating vaccine pages	Basic digital literacy skills, UI design skills	This involved simply adding more pages the same way as the previous ones were added and pasting on them the data collected in the research steps. However, since these pages were to be accessed by the filter system, they were added as pages called ‘posts,’ which are accessible by the search and filter plugin. Since this step involved working with the deeper functions of Wordpress, I took major help from my mentor here.
Organising vaccine ages into filtering categories	Secondary research skills, Basic digital literacy skills	First I researched all the different ways vaccines can be segregated and divided, in order to make sure that if there are more ways commonly used to divide vaccines other than the ones mentioned in the specifications (age groups, extent of mandation), then I would know them. I then made many simple google searches to identify in which age groups, necessities and disease type each vaccine fits in. Since the vaccine pages were added to be accessible via search and filters, these pages opened a side bar which displayed a checklist for the various categories they were included in. Via said checklist and the research done prior, I organised the vaccines into categories. Since this step involved working across many pages and the deeper functions of Wordpress, I took major help from my mentor here.

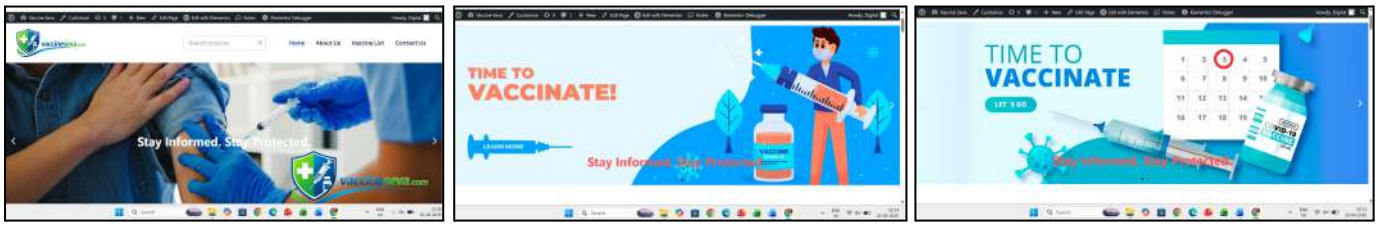
The Solution

https://vaccineseva.com/

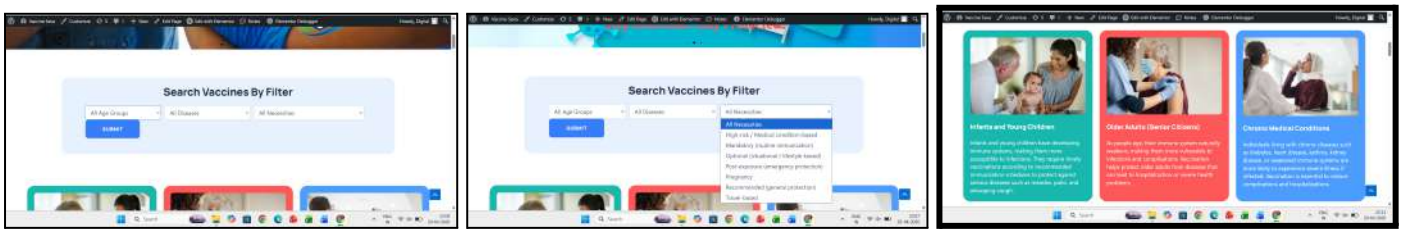
This is an educational website created to simplify vaccine information and spread awareness about vaccination. It provides organised information about different vaccines, including their uses, age groups, benefits, and diseases prevented. The website also includes filters and myths vs facts sections to help users easily find information and clear common misconceptions about vaccines.

Home page

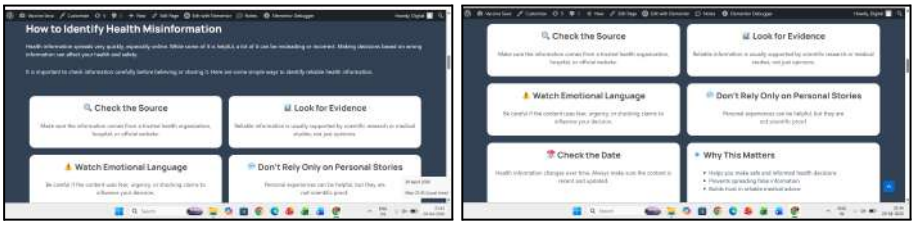
🌐 **Home page header section:**
The header includes links to all pages, excluding individual vaccine pages, followed by a slider that cycles between displaying three images, 5 seconds each, with the tagline ‘Stay Informed, Stay Protected’, which reflects the key idea of my solution, of people being able to take better medical decisions in terms of vaccination if they have the right information.



🌐 **Hero section: Filter section:**
This section includes three dropdowns (age group, disease, necessity) on which the user can fill information appropriate to the vaccines they want to know about. After clicking the submit button they will see a list of vaccines based on the options they selected.



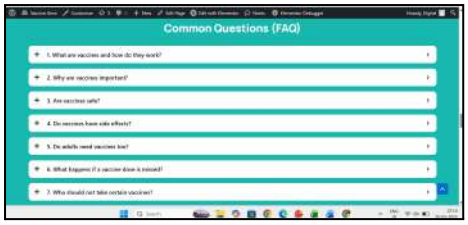
🌐 **Content section: Misinformation identification section:**
This section describes some ways to easily tell apart credible information from misinformation.

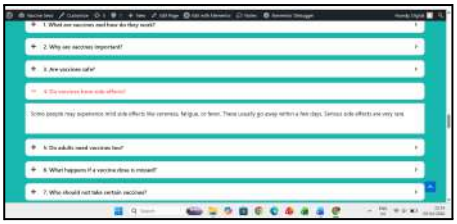


🌐 **Content section: Myths vs Facts section:**
This section features several cards with the most common vaccination myths on them. Hovering on them shows the actual facts behind those myths.



🌐 **Content section: Common Questions section:**
This section shows some of the most common questions people have about vaccination. Clicking on a question opens a dropdown showing the answer.






🌐 **Content section: Vulnerable Groups section:**
This section lists various vulnerable groups (people with diabetes, heart disease, etc) who may have different vaccination needs. Clicking on the name of the group opens a dropdown specifying the group's respective vaccination needs.





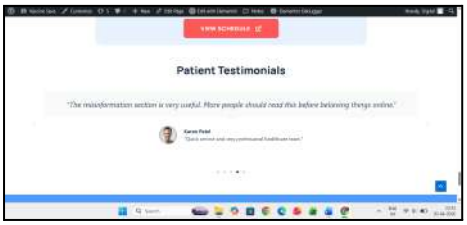


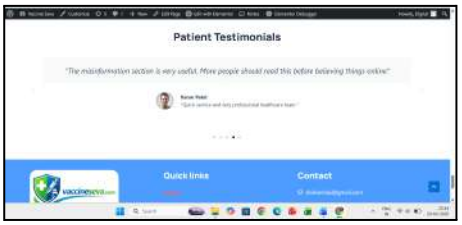
🌐 **Content section: Vaccination Schedule section:**
This section provides a link to the vaccination schedule pdf provided by the Indian government, along with a visual infographic of the schedule also provided by the government.

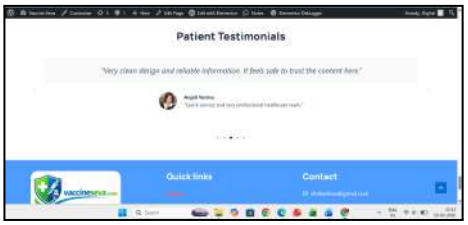





🌐 **Footer section: website testimonials and Disclaimer**





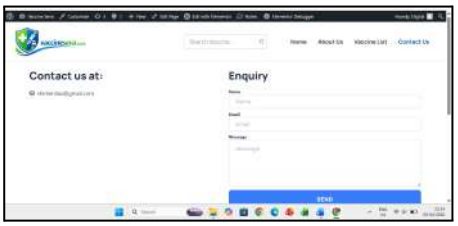
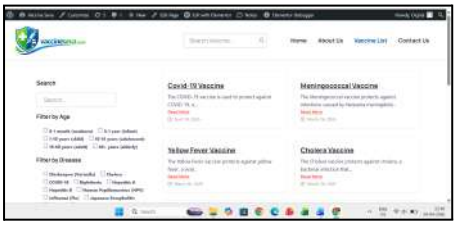
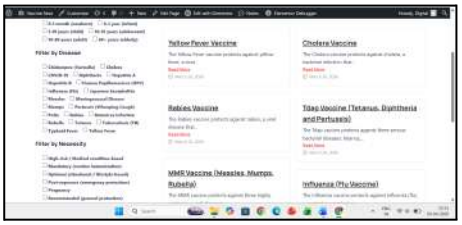


About us page

This page gives an introduction of the website and its content.





Contact us page
This page provides an email address for users to contact me.

Vaccine list page
  

Additions to the Specifications	
No specifications from the original plan were removed or changed for the final design. Specifications were only added. Below are specifications for elements of the website added during development along with justification for their inclusion.	
Addition	**Justification**
☑Vaccines for vulnerable groups must be separately mentioned.	People with special medical conditions such as diabetes, heart disease, etc, are more vulnerable to certain diseases, and are particularly advised to take certain vaccines. (NFID) Because VaccineSeva.in is to be a complete resource for information on vaccines, this was a crucial part of the website that the original specifications did not include.
☑Ways to identify misinformation must be mentioned alongside misinformation.	Though the initial specifications included the mention of common misconceptions in several places, people not being able to tell apart credible and non-credible information (or misinformation) is a large cause for the spread of misinformation. (Munusamy et al.) Hence, this addition was made.
☑Real user testimonials must be mentioned.	Adding real user-testimonials/reviews helps build the user's trust in a website, as user-testimonials act as social proof and help dispel skepticism. (Adams) Gaining the trust of its users is crucial, as the VaccineSeva.in aims to provide the user with the most complete and correct information.

SECTION D: EVALUATING:

Specification-Based Testing		
Below are results of the tests specified in the success criteria for the design specifications.		
Tests	**Success**	**Evidence**

The website must have arial as a universal font.	✅	My website uses the default arial font on all of its pages.
A design philosophy expert must confirm that the colours evoke the same emotions as I claim.	✅	A design philosophy expert indeed confirmed that the website invokes the same emotions of professionality and credibility as I explained in the specification. He further added that the use of different colours for certain sections such as ‘myths vs facts’ and ‘vaccines for special categories’ was a good choice as the same colours scheme on such a long home page would have looked boring.
The language must be understandable to my target audience, Indian citizens, and should also be approved by a medical professional as adequate.	✅	Medical communication websites aimed at a general target audience should avoid using technical, medical-specific vocabulary wherever possible and substitute them for more common vocabulary. Wherever the use of scientific vocabulary is absolutely necessary, such as the name of a disease or condition, the work should be immediately followed by a definition that would explain the term to a common audience.
All sources used must be cited.	✅	Sources used are cited on the footers of all vaccine pages.
Each page must have the last update date mentioned.	✅	The date (month/year) of the last update is mentioned on the footers of all vaccine pages.
The disclaimer must clearly express the idea that this website cannot replace professional advice or diagnosis.	✅	The footer of the main page has the disclaimer: “This website is created for educational and informational purposes only. It is not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider for medical concerns.” Meanwhile, the About Us page has the disclaimer: “Vaccine Seva is an educational initiative and does not replace professional medical advice. Users are encouraged to consult qualified healthcare professionals for medical decisions.” Both disclaimers use clear language, such as ‘educational and informational purposes only’ and ‘It is not a substitute for.’ This emphasizes the idea of using the website only for information, but still consulting a medical professional for advice or diagnosis before taking any medical steps.
The search bar must be able to show the closest results when the name of a vaccine or disease is searched.	✅	The search bar shows all the closest vaccine page results whenever something. Clicking on the results leads the user to the respective vaccine page.
The navigation must work as intended and show a list of vaccines that fit the information entered (filters applied).	✅	As seen in the recording below, entering different requirements into the filter system yields different
The list of vaccines is approved by a medical professional as complete.	✅	I showed my website to a doctor, who approved the content on my website, including the list of vaccines.
The information must be sourced from credible sources such as government or medical	✅	All information from the website has been taken from credible sources, such as government websites and well-known medical communication websites.

websites.		

Survey

I created and sent a questionnaire to my target audience, Indian citizens and caregivers, which would investigate and measure the effectiveness of the various features of VaccineSeva.com.

Questions

Do you think VaccineSeva.com acts as an organised and trustworthy database for information on vaccines and clears doubts and misconceptions about the same.

- ◯ Yes
- ◯ No

If yes/no, why so?

Short-answer text

Did you think you can gain any value in your knowledge or thoughts about vaccination by using VaccineSeva.com?

- ◯ Yes
- ◯ No

Rate the information on the main page based on usefulness ona scale of 5.

	1	2	3	4	5	
Not Useful	◯	◯	◯	◯	◯	Very Useful

Rate the information on the pages for the individual vaccines on a scale of 5

	1	2	3	4	5	
Not Useful	◯	◯	◯	◯	◯	Very Useful

Rate my website in terms of ease of navigation. (While navigating vaccines or trying to find specific information)

	1	2	3	4	5	
Difficult to Navigate	◯	◯	◯	◯	◯	Easy to Navigate

How helpful was the filter system for finding desired pieces of information on a scale of 5?

	1	2	3	4	5	
Not Helpful	○	○	○	○	○	Very Helpful

Which feature of VaccineSeva.com help you the most?

Short-answer text

What feedback do you have to improve VaccineSeva.com

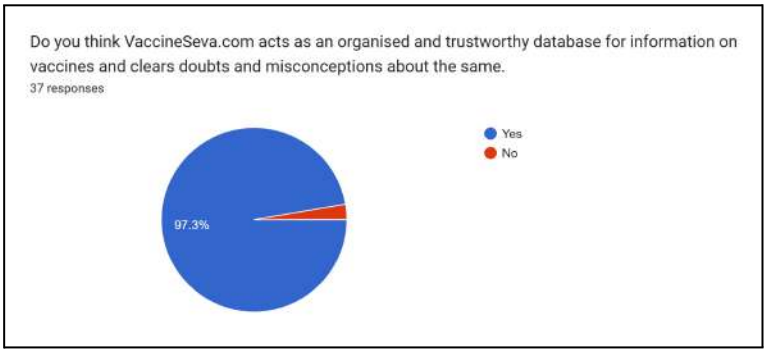
Short-answer text

Would you reccomend VaccineSeva.com to others?

- ○ Yes
- ○ No

Survey Results

👉 Below are the results of the above questionnaire. Responses to multi-choice or scale-selection questions are presented in graphs, while for text-response questions, all unique responses are presented.



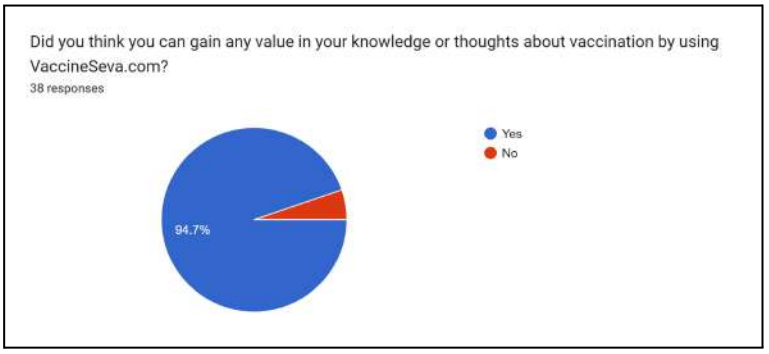
👉 If yes, why so?

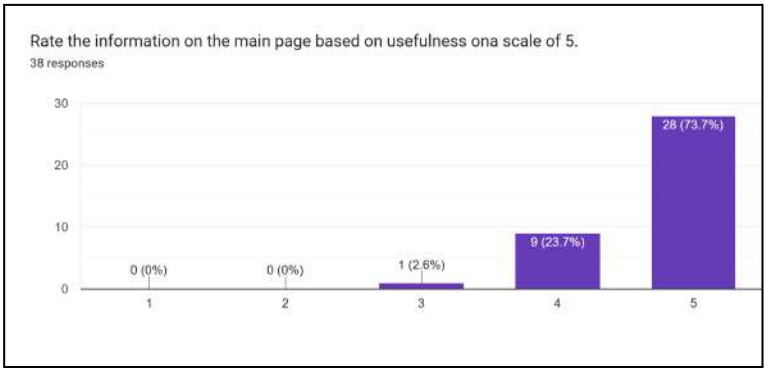
- ➢ The website is very clear, well organised, and easy to understand without being overwhelming.
- ➢ The website is highly informative and helped improve my understanding of vaccines.
- ➢ The filters and navigation system make it easy to quickly find the required information, making the overall experience smooth and user-friendly.
- ➢ The information appears scientifically accurate and trustworthy, especially because it is explained clearly and cites its sources.
- ➢ The website looks professional, useful, and thoughtfully designed. The simple layout and focused approach

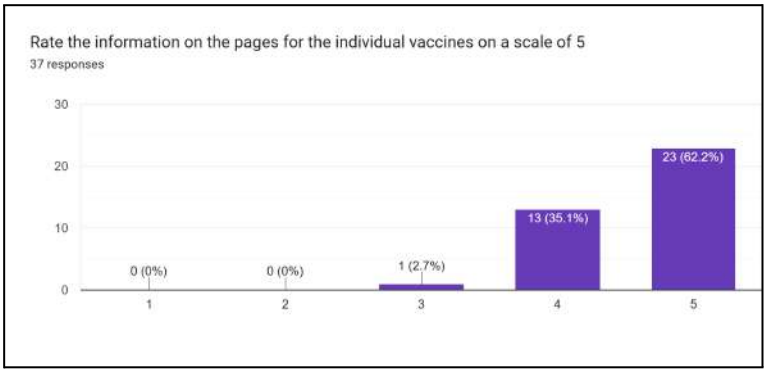
make it very impressive.

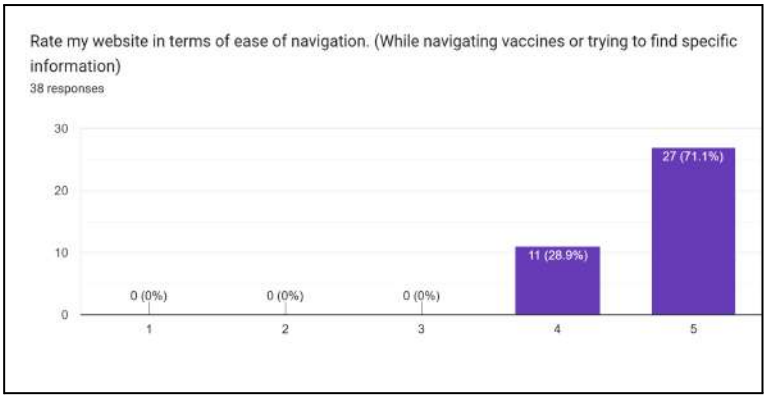
👉 If not, why so?

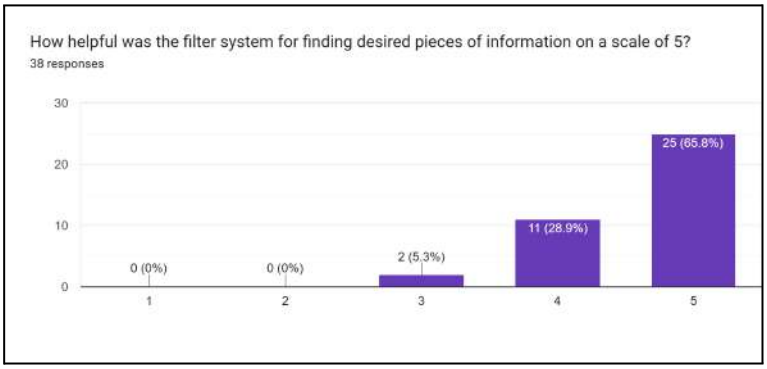
- ➢ The website needs an explicit link to a credible source or organisation in order to seem trustworthy.
- ➢ The website must account for recent studies that show inconclusive data on some vaccines, resulting in gaps in definite information.
- ➢ The website would need to be continuously updated as new information on vaccines becomes available.









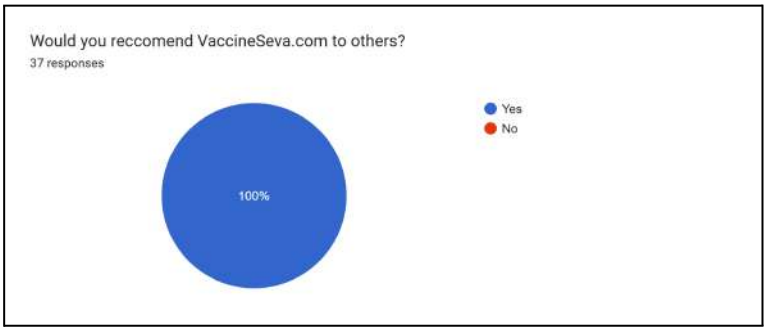


👉 Which feature of VaccineSeva.com helped you the most?

- ➢ All features were mentioned by various respondents as being the most helpful.

👉 What feedback do you have to improve VaccineSeva.com?

- ➢ It would be helpful if you add other languages, especially Hindi and Marathi, as a large part of the population is more comfortable with them.
- ➢ Travel vaccines are often difficult to find. Hence, for them you could mention particular places where they can be found.
- ➢ A filter should be added to direct to travel vaccines based on specific countries.
- ➢ Some important vaccines that are still missing from the database must be added.
- ➢ The spectrum of categories could be broadened with categories such as pet-related vaccines.
- ➢ While the filter system is useful, it should be explained beforehand so that the user immediately understands how to use it.



Modifications to Design

Based on the results of the tests, especially the feedback received via the survey, below are modifications that need to

be made to VaccineSeva.com.

Additional Specification	Justification
👉 The website must allow the user to switch between Hindi, English and regional languages (marathi, gujarati, etc).	Several respondents of the survey gave the suggestion of making the website multilingual due to the much of my target audience preferring hindi or regional languages over english. Only ⅛ people in India can speak english. (Azam et al.) Therefore, since my target audience spans throughout India, including Hindi and regional language modes in my website is necessary.
👉 The home page must include a text directly above the filter selection section explaining how the filter system works.	The filter section is crucial to communicating information of individual vaccines with the user. A respondent to the survey said that the navigation may be unclear. Therefore, adding clarity to the navigation is necessary.
👉 The website should have information on all vaccines currently administered in India.	My initial research of vaccines sought to cover all vaccines in a few categories such as those part of the universal immunization program, travel vaccines and certain other categories that I thought would cover all vaccines. However, as pointed out by some respondents (many of whom were doctors), some vaccines seem to be missing from the list. Therefore, expanding the specification to 'all vaccines currently administered in India' was necessary in order to make a truly complete list of vaccines.
👉 The category for travel vaccines must be further divided into vaccines for different travel destinations (countries).	Travelling to different countries requires different vaccines, though on the website they are all grouped under a single category. Based on a respondent's feedback, for a traveler looking for vaccines to take for a particular journey, it would be inconvenient to go through all the travel vaccines. Therefore, to keep the navigation efficient, it is necessary to expand the travel categories of vaccines into further categories based on travel destinations.
👉 The pages for travel vaccines must include a list of authorised clinics that can administer said vaccine according to the user's region.	A respondent, who is my relative, a doctor and a parent, gave the suggestion of adding information to help the user find authorised clinics for travel vaccines, as they are not as widely available as other vaccines. Some travel vaccines such as the yellow fever vaccine can only be administered by authorised clinics. (CDC) Therefore, in order to make my website a complete database for vaccines, implementing this suggestion is necessary.
👉 The vaccine categories must include the additional category of pet vaccines, further divided into vaccines for types of pets.	A respondent suggested the inclusion of vaccines for pets. This would make the information on my website more complete. Further dividing this category into vaccine categories for types of pets would make navigation easier. Therefore, the addition of categories for pet vaccines is necessary.

The Impact of VaccineSeva.com

In the below section I will analyse the success of my solution with all improvements in mind, keeping in mind the original problem statement and design brief.

Impact on Target Audience

The target audience of Indian citizens and caretakers faced the following problems: (Summary of Design Brief)

- ➢ The lack of a credible source with complete information on vaccination.
  - My research suggests that most if not all negative information about vaccines comes not from credible websites or doctors, but from social media.

- ➢ Being discouraged from vaccination by myths and misconceptions.
  - 25% of Indians report having heard negative misconceptions about vaccines. 37.5%of people surveyed have been discouraged from vaccination due to hearing concerning information.
- ➢ Being at risk of diseases due to being unvaccinated.
  - 500,000 people die per year in India due to diseases that vaccination could have prevented.
- ➢ People have made the internet rather than doctors their primary source of medical information, endangering them.
  - Research suggests that ever since the internet became accessible in the 21st century, people are steadily replacing medical professionals with it, increasing the likelihood of misinformation spreading.

VaccineSeva.com solves this problem by:

- ➢ Giving the audience a website that sources its information from credible sources and provides a wide range of information about a wide range of general topics regarding vaccination, as well as specific vaccines.
  - 97.3% of people surveyed agreed that VaccineSeva.com serves the above mentioned functions.
- ➢ Making it easy for the audience to find any information regarding vaccines they aim to find in a single website.
  - 71.1% of people surveyed rated the navigation (filter and search) system a 5/5, while the rest 28.9% rated it 4/5.
- ➢ Helping the audience differentiate between common myths and facts.
  - All common myths or questions that came up in either primary or secondary research have been addressed on VaccineSeva.com.
- ➢ Telling the audience about the benefits of vaccination and encouraging them towards it.
  - The answers for said common questions and myths have been framed to constantly mention the importance of vaccination and its benefits.
- ➢ Answering all their vaccination doubts on a single website.
  - 94.7% of people surveyed said that the information provided on VaccineSeva.com will add value to their thoughts and ideas about vaccination

Potential Negative Effects

As mentioned several times throughout, there are constantly new developments in the realm of vaccines. Hence, if VaccineSeva is not constantly updated with every new piece of vaccine information, it may become outdated and lose its credibility. This highlights the problem of having to constantly check vaccine information updates and update the website accordingly, the solution for which can be found with another design cycle.
This can, however, be presently mitigated with attentive management of the website and constant updates. It can be integrated with an AI that can automatically update any new developments about vaccination.

Resolvement of Problem Statement

As demonstrated by the above two sections, the solution solves all the problems faced by the target audience, the gravity of which was shown by prior research, but at the same time has a small side effect that is avoidable. Therefore, its accomplishments greatly outweigh its single flaw.

Therefore the problem initially stated: Indian citizens having limited information regarding vaccination policies and regulations, negatively affecting the health of the population, can be solved by VaccineSeva.com.

Thank You

Appendix A: Citations

Criteria A

Works Cited

Aggarwal, Sumit, et al. “COVID-19 Vaccine Hesitancy among Adults in India: A Primary Study Based on Health Behavior Theories and 5C Psychological Antecedents Model.” *PloS One*, vol. 19, no. 5, Public Library of Science, May 2024, pp. e0294480–80, https://doi.org/10.1371/journal.pone.0294480.

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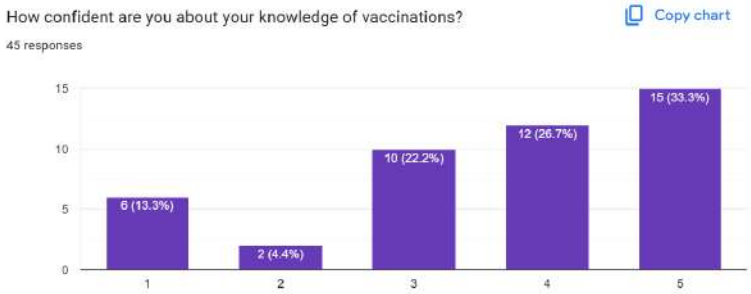
Appendix B: Research Survey Results

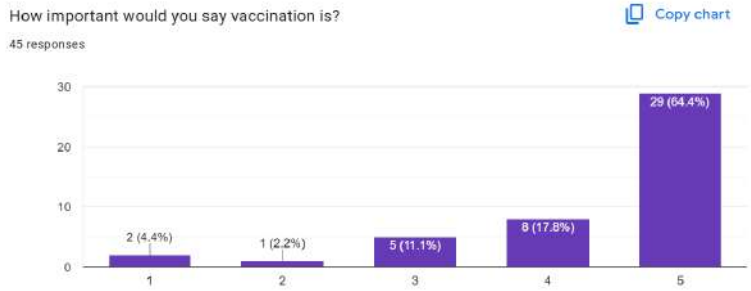
Below are the questions, results and other details of the survey. These are presented as an appendix as not all of the results may be relevant to understanding the overall research.

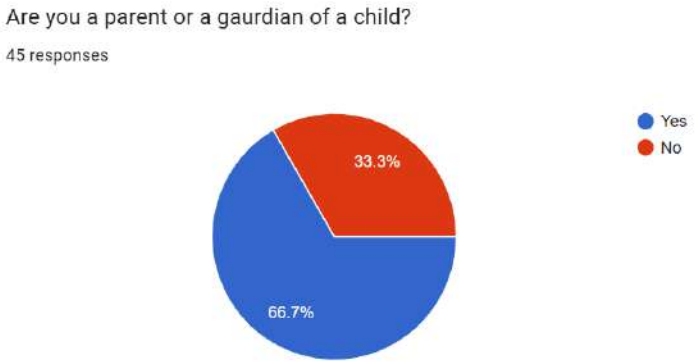
Details

Link: Vaccine Awareness
Title: Vaccine Awareness
Sent to: School learners, parents and relatives

Questions and Responses:

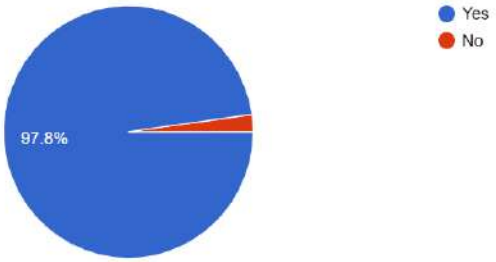






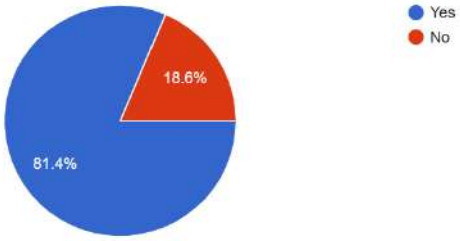
Have you or your family recieved vaccinations in the past?

45 responses



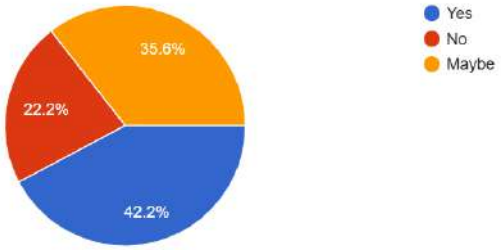
If there are children in your family, are they up to date with their vaccinations?

43 responses



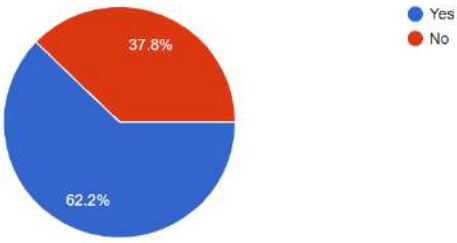
Do you know which vaccines are mandatory in India?

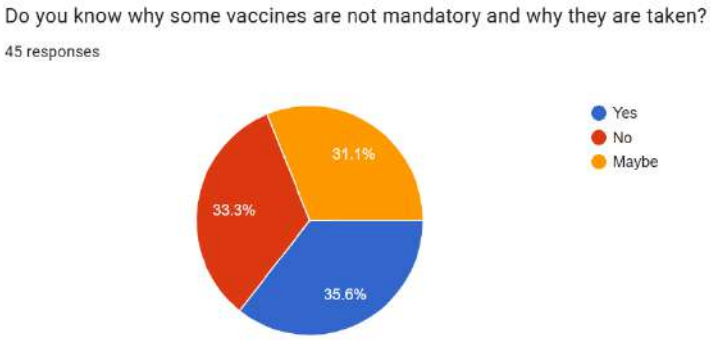
45 responses

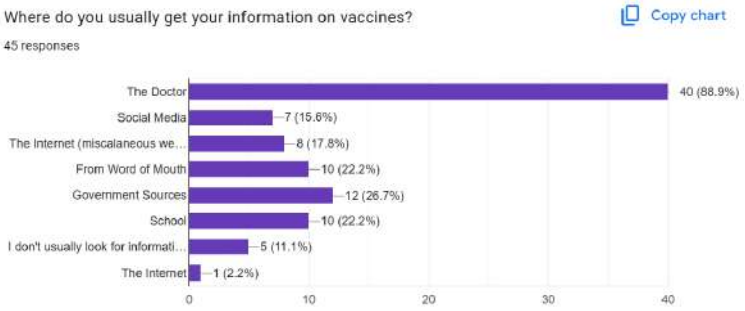


Do you know there are different vaccines reccomended at different ages other then the basic ones given to younger children?

45 responses







Note: 'Internet' shows on the graph twice due to an error caused by the change of the option from 'internet' to 'internet (miscellaneous websites)'. The combined percentage of both 'internet' options is 20%.

What challenges have you encountered with any of these sources of information?

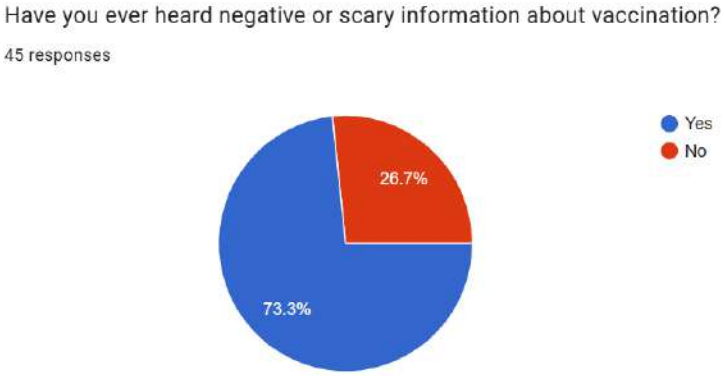
This was an open-ended question where respondents could answer in a short text. All responses can be summed up to:

- ➢ Sources provide information that is too technical.
- ➢ Sources give contradictory information.
- ➢ The internet/Social media is filled with misinformation.
- ➢ It is difficult to find updated information.
- ➢ Not much information where people should be getting it, such as government sites.
- ➢ Lack of clarity about side-effects.
- ➢ No challenges faced.

Do you still have questions about vaccines that have gone unanswered?

This was an open-ended question where respondents could answer in a short text. All responses can be summed up to:

- ➢ What are the long-term effects of vaccines?
- ➢ How are vaccines updated to combat new variants of diseases?
- ➢ How likely are you to get a disease if you don't get the vaccine?
- ➢ How long do vaccines last?
- ➢ What are the side-effects of vaccines?
- ➢ Why do some people see vaccines as bad despite them saving countless lives?
- ➢ What makes some vaccines mandatory?
- ➢ No questions that have gone unanswered.

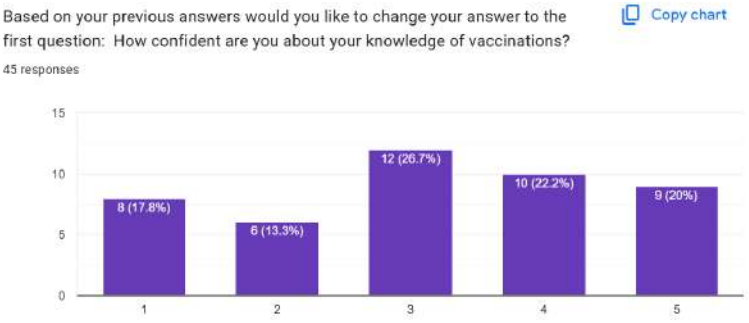


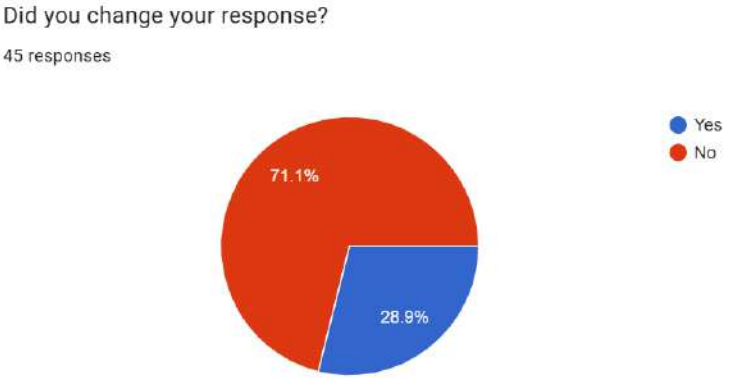
If you have heard negative or scary information about vaccines, where and how did you find it?

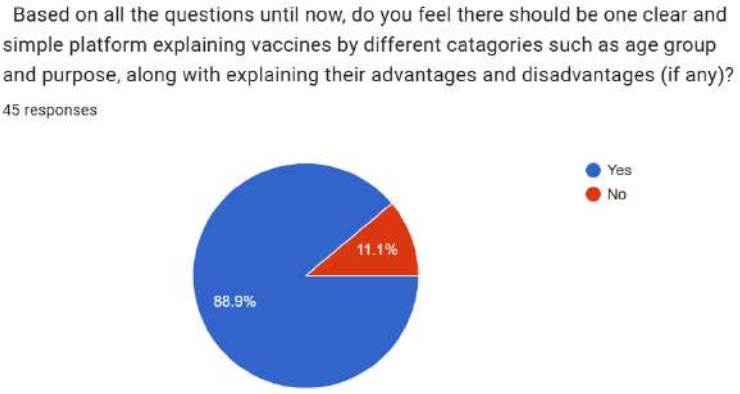
This was an open-ended question where respondents could answer in a short text. All responses can be summed up to:

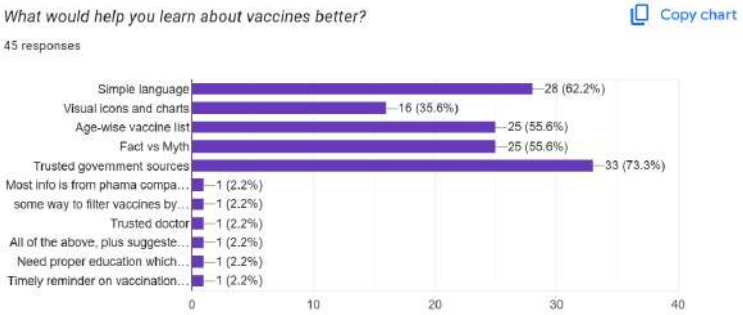
- ➢ Social media: Most common answers were related to social media
- ➢ Family/Friends/Word of mouth: Second most common answers were related to word of mouth
- ➢ Doctors





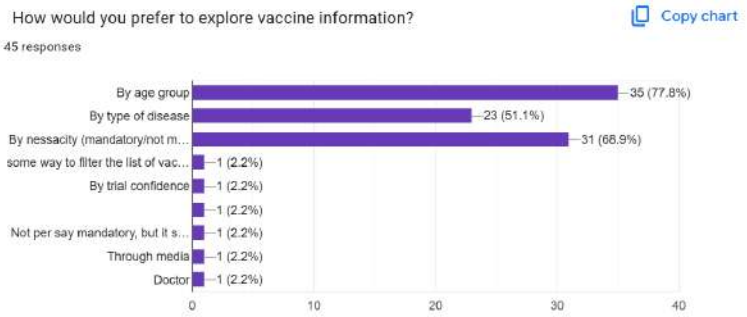






Options were:

- ➢ Simple Language
- ➢ Visual Icons and Charts
- ➢ Age-wise Vaccine List
- ➢ Fact vs Myth Clarification
- ➢ Other



Options were:

- By Age Group
- By type of Disease
- By Nessacity (Mandatory, Not Mandatory, Conditional)
- Other

